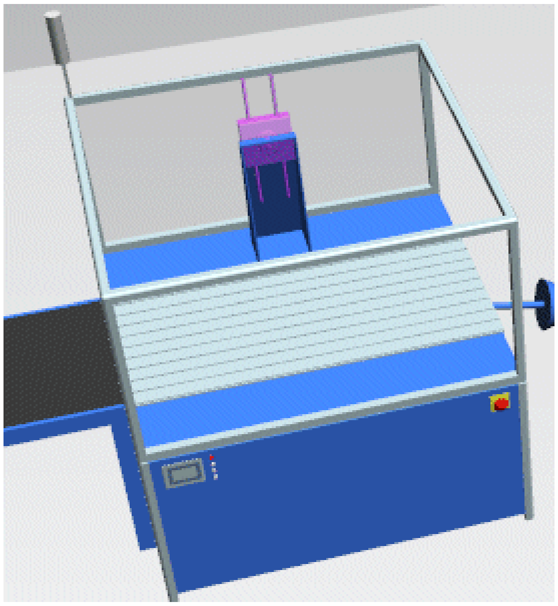
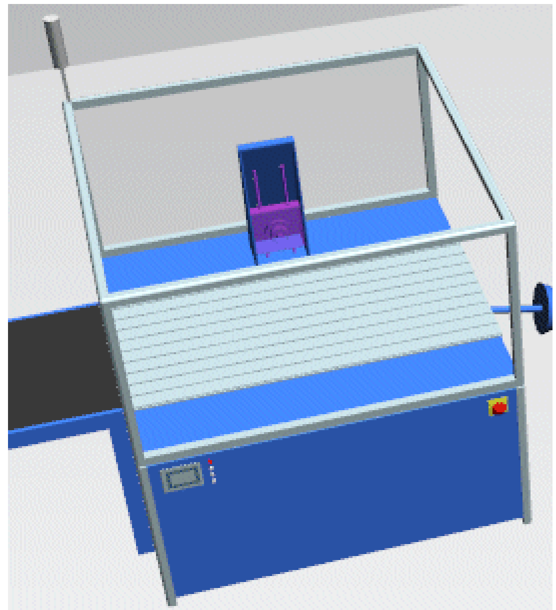
In the example below we showed **Pose1**. It moves the **Prismatic Joint** of the **ToolHolder** up when you press the **right arrow key**.





Compare Tab Poses and Tab Joint

Compare Configuring the Robot

Back to Model Joints and Poses

Video on YouTube

https://youtu.be/wfVN-mcWNsc?si=LepG6cgh0MDLKiIQ&t=22

Creating a Realistic Looking Model

Create a Realistic Looking Model

You can create a visually pleasing and realistic looking model.

You can:

- Insert Factory Walls and Create Passageways
- Add Text and Display Boards
- Add a JT Layout File to Your Simulation Model
- Add a Point Cloud
- Import JT Graphics Representing an Object

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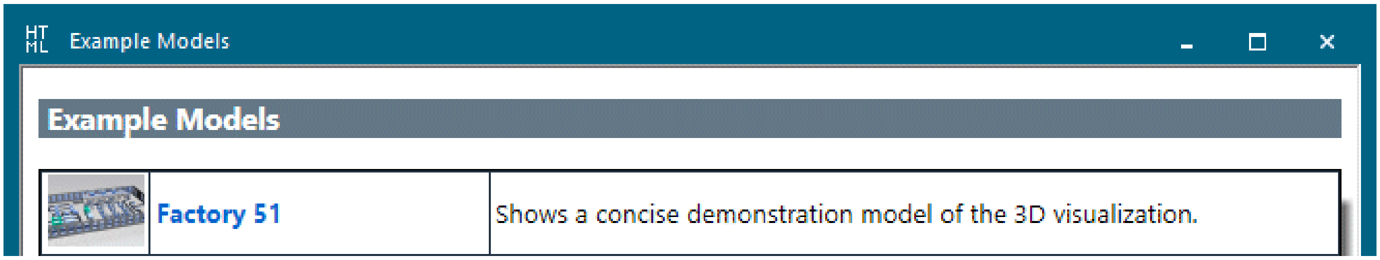
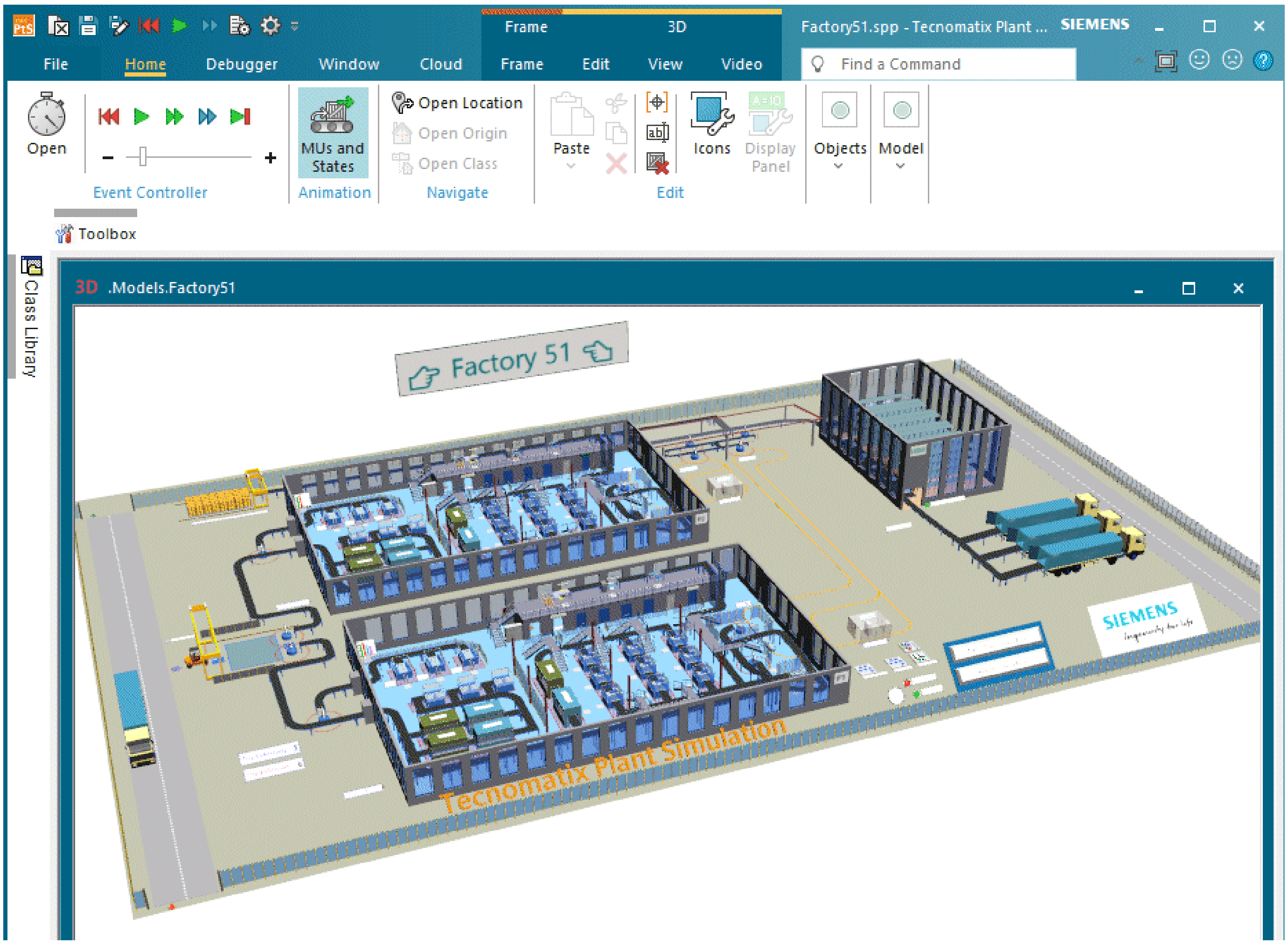
The model **Factory 51**, which you can open from the **Start Page** under **Example Models**, exemplifies these and other techniques.





Back to Visualizing the Material Flow

Enhance the Visualization of Your Simulation Model in Omniverse

You can use the **Omniverse Connector** to create an good looking visualization of your *Plant Simulation 2606* simulation model in Omniverse, compare these videos:

https://support.sw.siemens.com/en-US/okba/KB000179528_EN_US

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https://support.sw.siemens.com/en-US/knowledge-base/KB000179496_EN_US

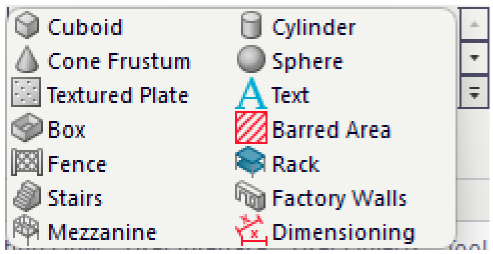
Insert Factory Walls and Create Passageways

Insert Factory Walls and Create Passageways

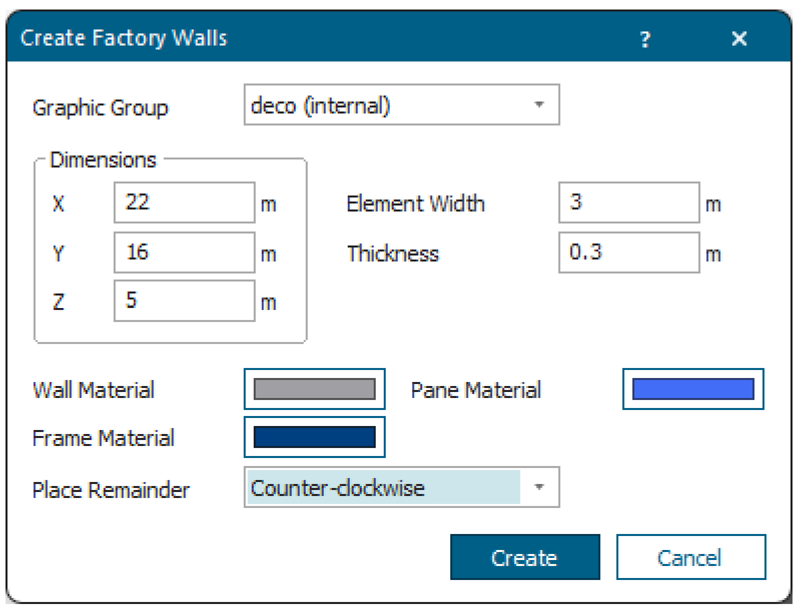
In our sample model we create four factory walls around a *conveyor*.

Proceed as follows:

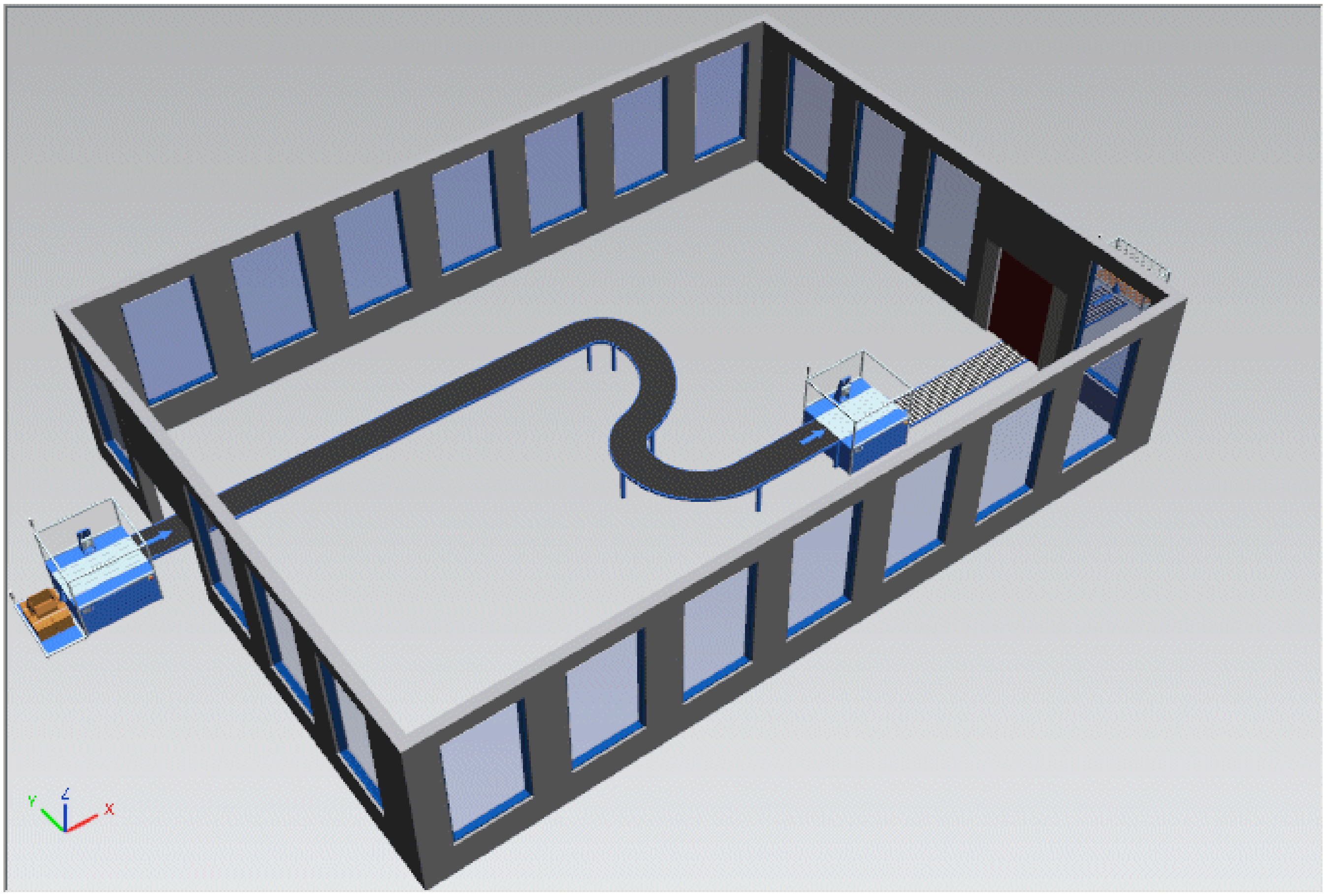
- Click the **Edit** ribbon tab in *3D*. Click **Edit > Insert Shape > Factory Walls**.



- Select the settings for the walls. We selected the settings below:



- The model then looks like this:



Now we would like to cut a passageway through the wall between the *Station* and the *Conveyor* so that we can extend the *conveyor* through the passageway.

We will demonstrate how to:

- **Cut an Open Passageway in a Wall**
- **Create a Passageway and a Gate**

Back to Create a Realistic Looking Model

Cut an Open Passageway in a Wall

You can cut an open passageway through a wall.

Proceed as follows:

- Click the mouse at the position in the factory wall into which you want to cut the passageway. We clicked the position directly above the *tooltip*.

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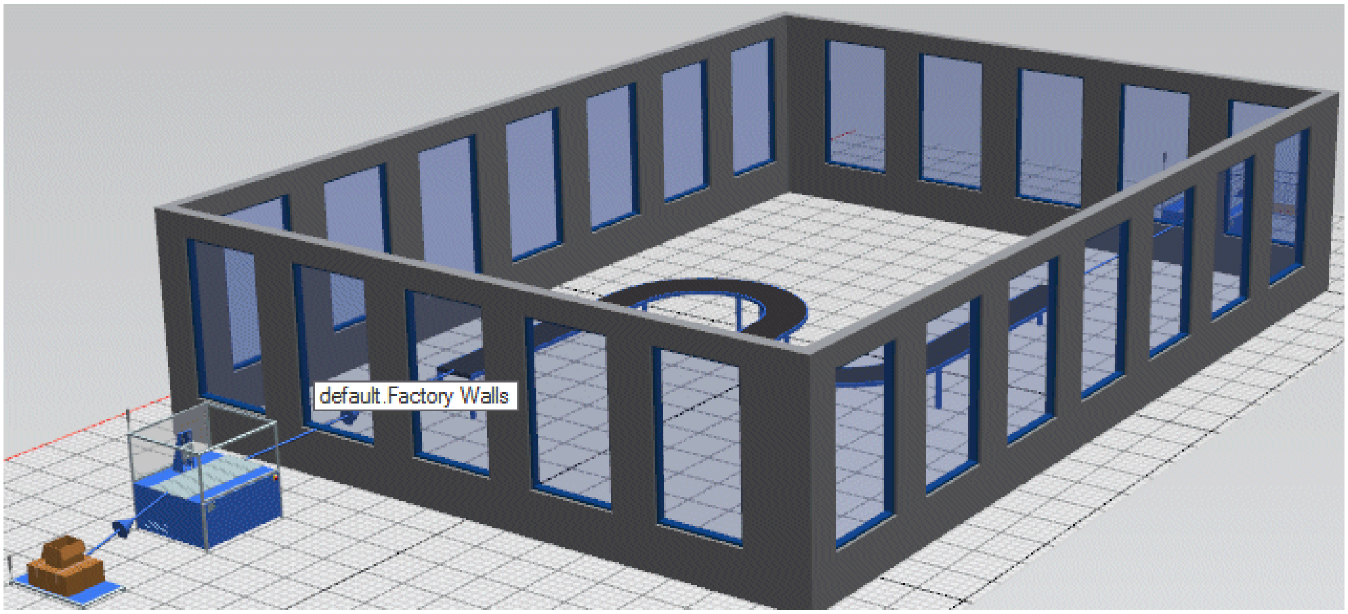
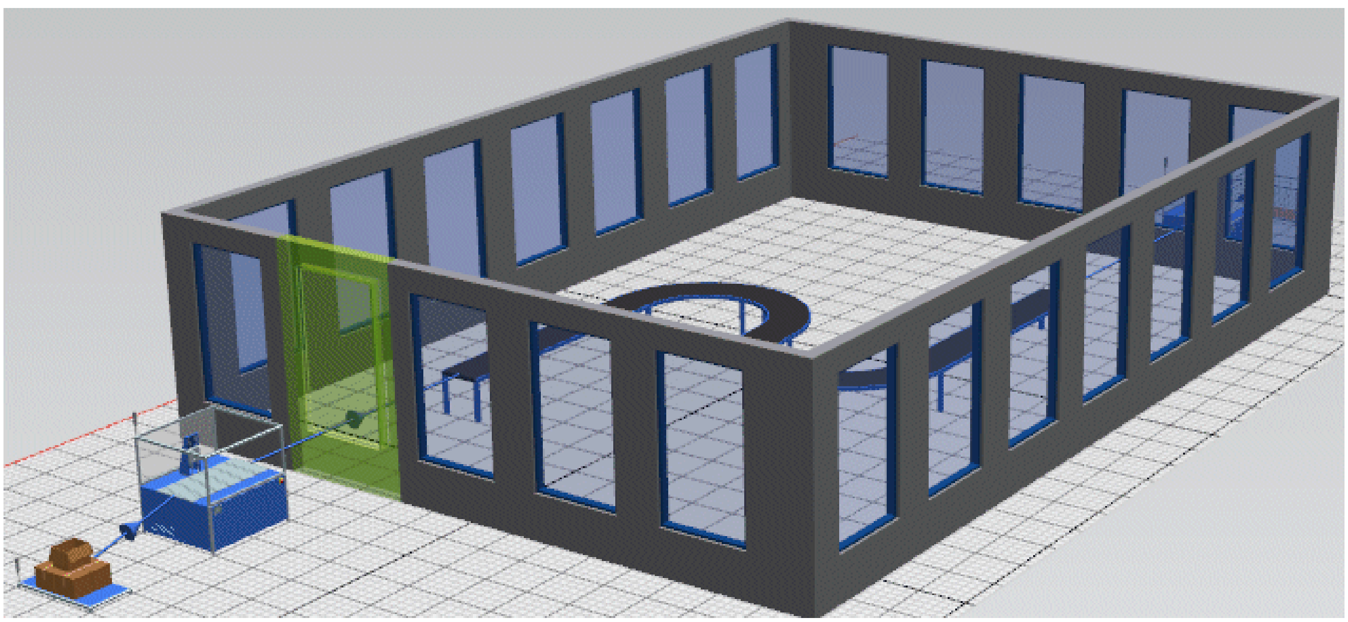


- Press the **+** key on the keyboard until only the segment is selected which you want to cut.



- Delete the segment (the graphic) by pressing **Del** on the keyboard. *Plant Simulation* then shows a message box. Click **Yes**.

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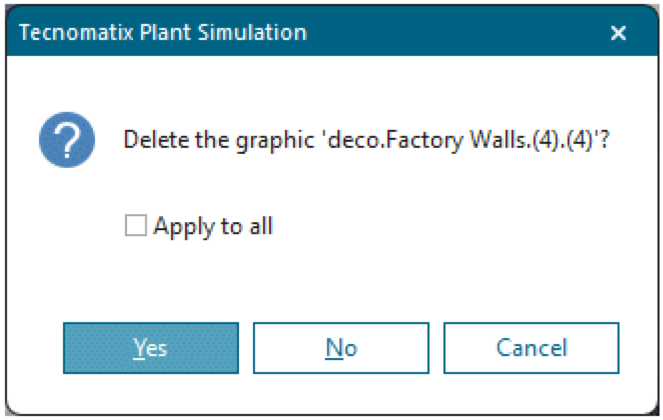
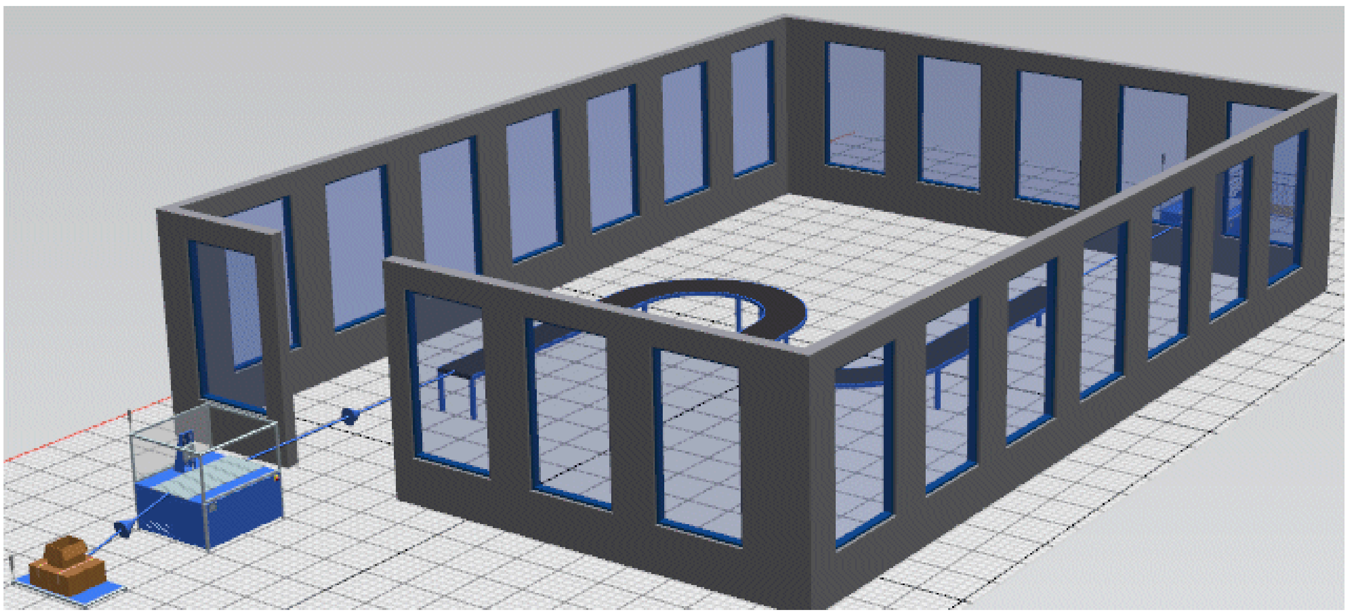
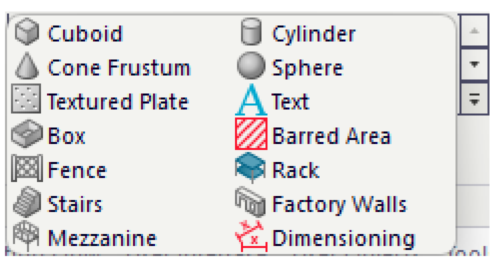


- The model now looks like this:



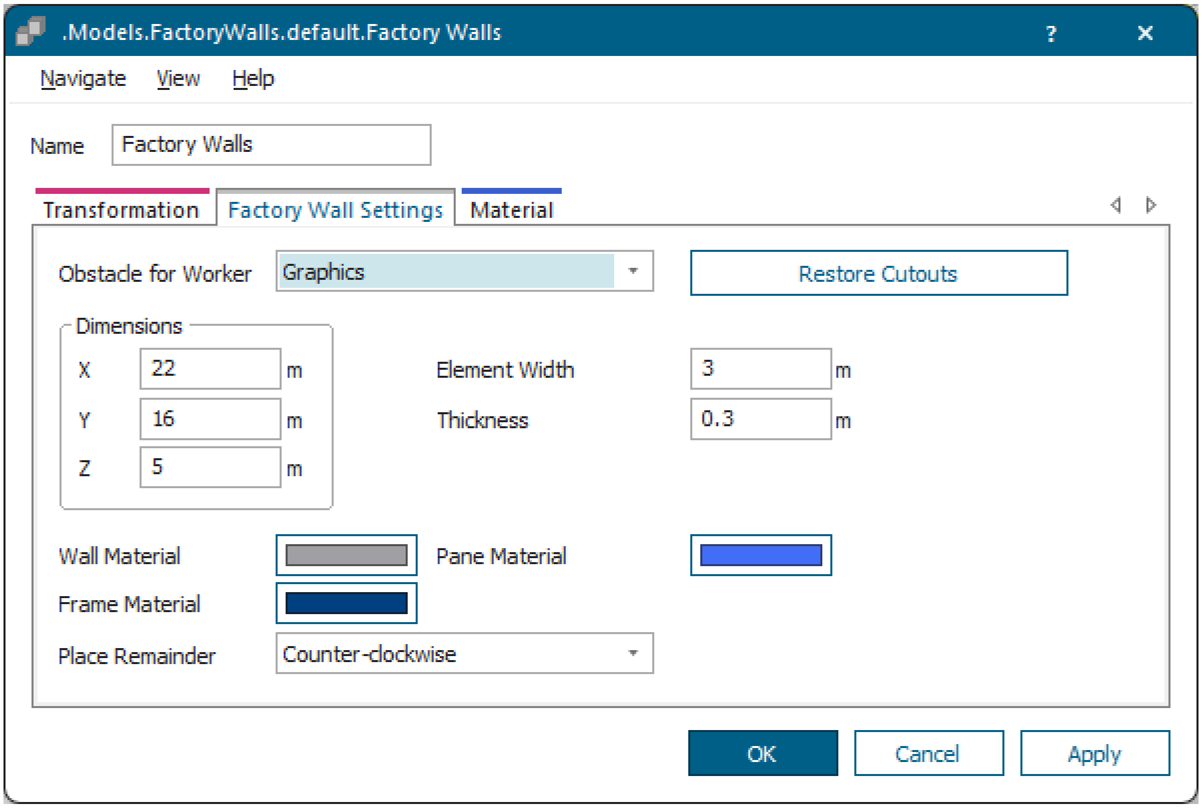
As we do not want the passageway to be open at the top, we add a lintel. To do so, proceed as follows:

- For the lintel we use a **Cube**. You could also use a **Textured Plate** though.

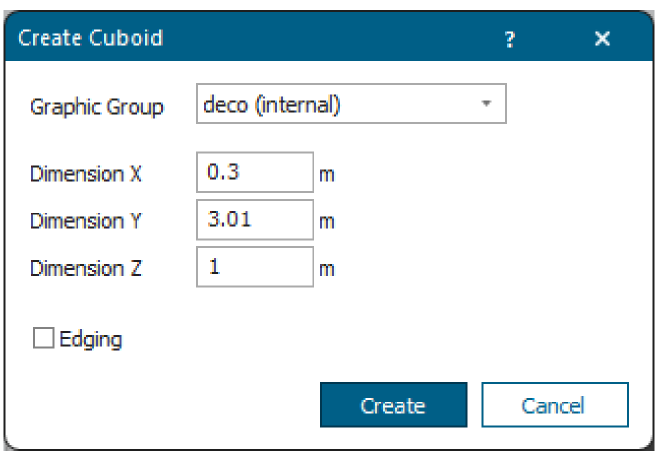


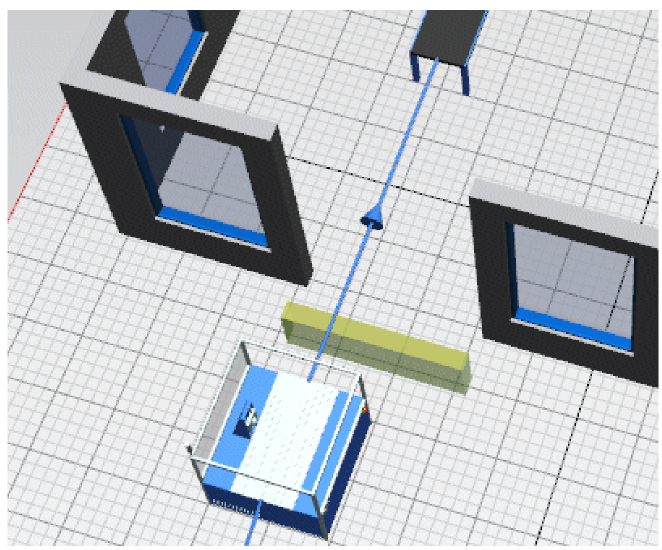
As we cannot change the **Thickness** of the cube any more after we inserted it, we open the **Factory Wall Settings** and note down the value of the **Thickness** of the factory walls, namely **0.3** meters. This way we can ensure that the lintel is just as thick as the factory wall.

Unpublished work. © 2026 Siemens



- Click the **Cube** and enter the settings below. The **Thickness** of `0.3` meters of the factory walls corresponds to the **Width** of 0.3 meters of the cube. Click **Create**, drag the mouse pointer to the position of the passageway at which you would like to insert the lintel and click the mouse button.





- Change to **Planning View** and place the lintel roughly with the arrow keys on the keyboard.
- As this is not accurate enough, click the lintel with the right mouse button and **Show** its **3D properties**. Click in the text box **X-Position** with the left mouse button and roll the mouse wheel until the position of the lintel looks correct in this direction.

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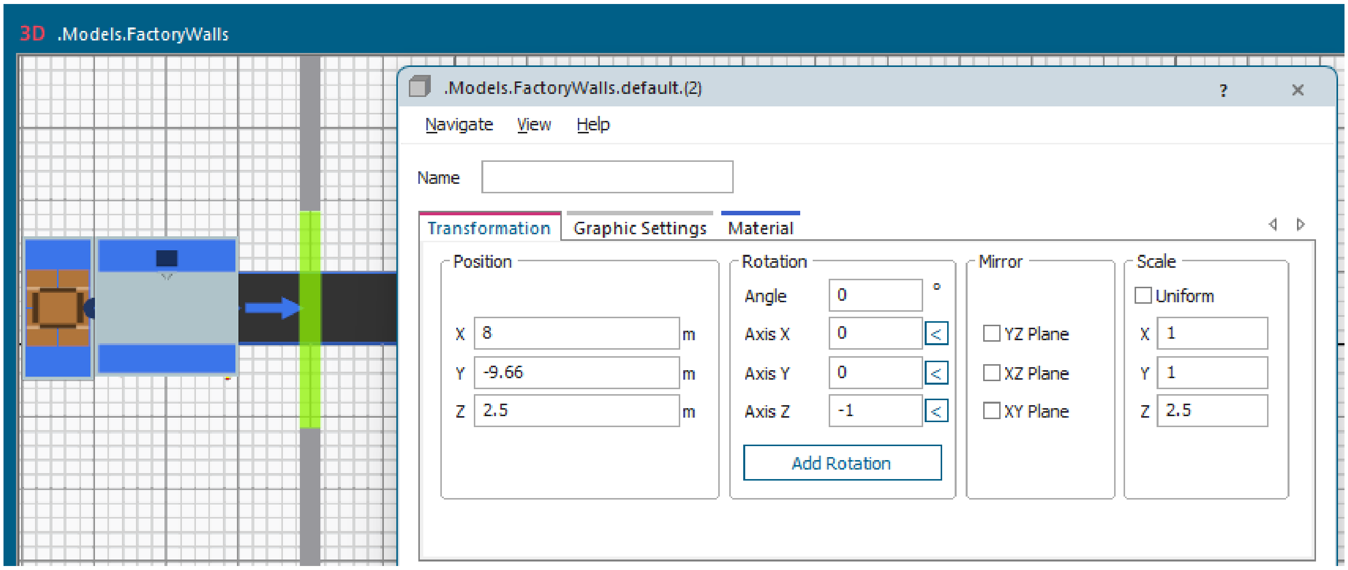
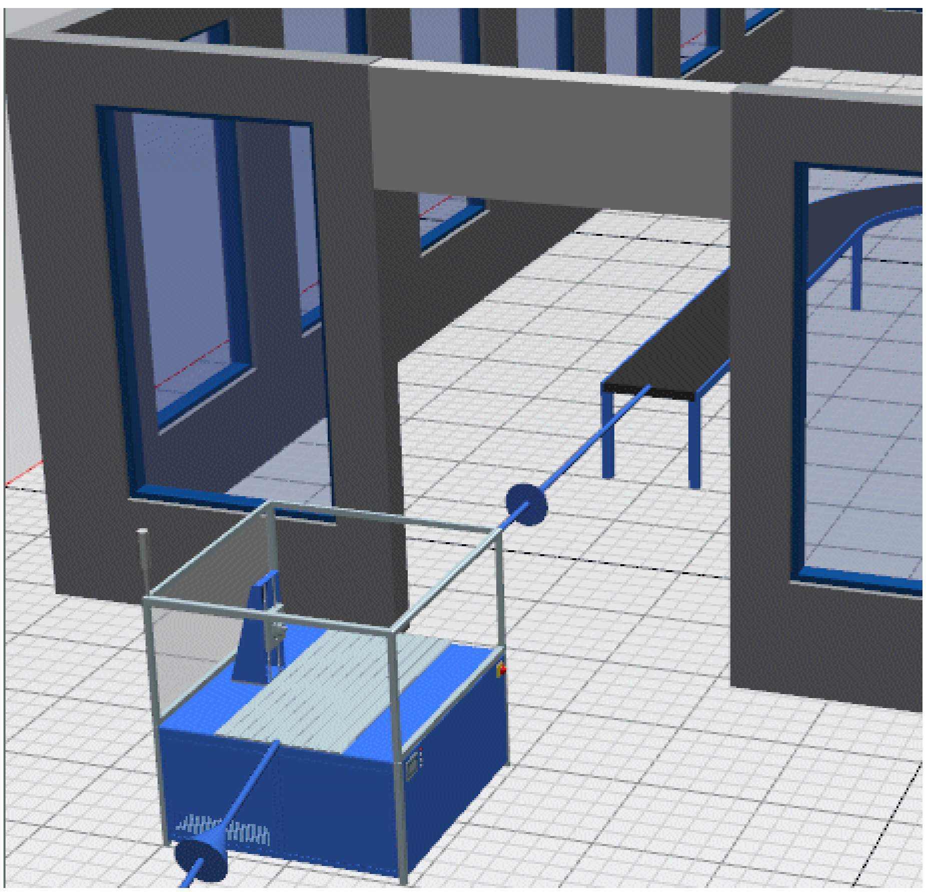


- Deactivate **Planning View** and adjust the **Z-Position** of the lintel until it looks correct. Proceed as you did for the **X-Position**.



- In the next step we just have to adjust the material/the color of the lintel so that it matches the material of the factory walls. Open the **Factory Wall Settings**. Click **Wall Material** and copy the material in the dialog that opens.

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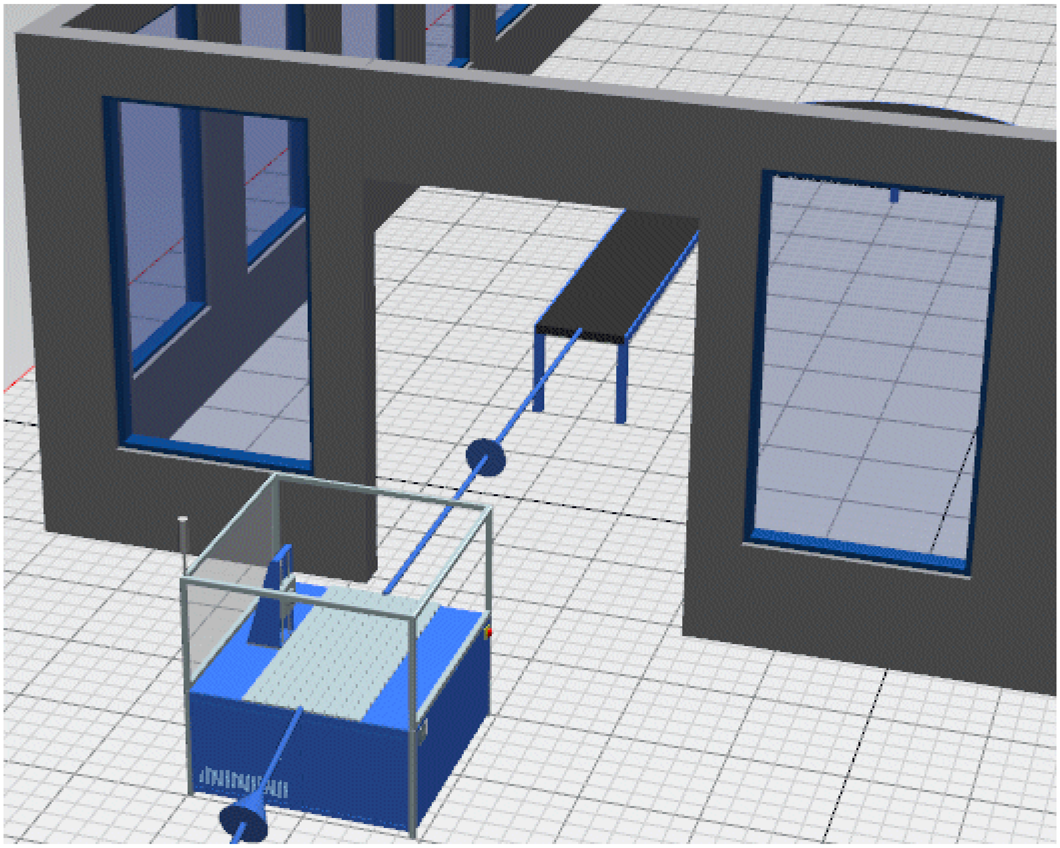
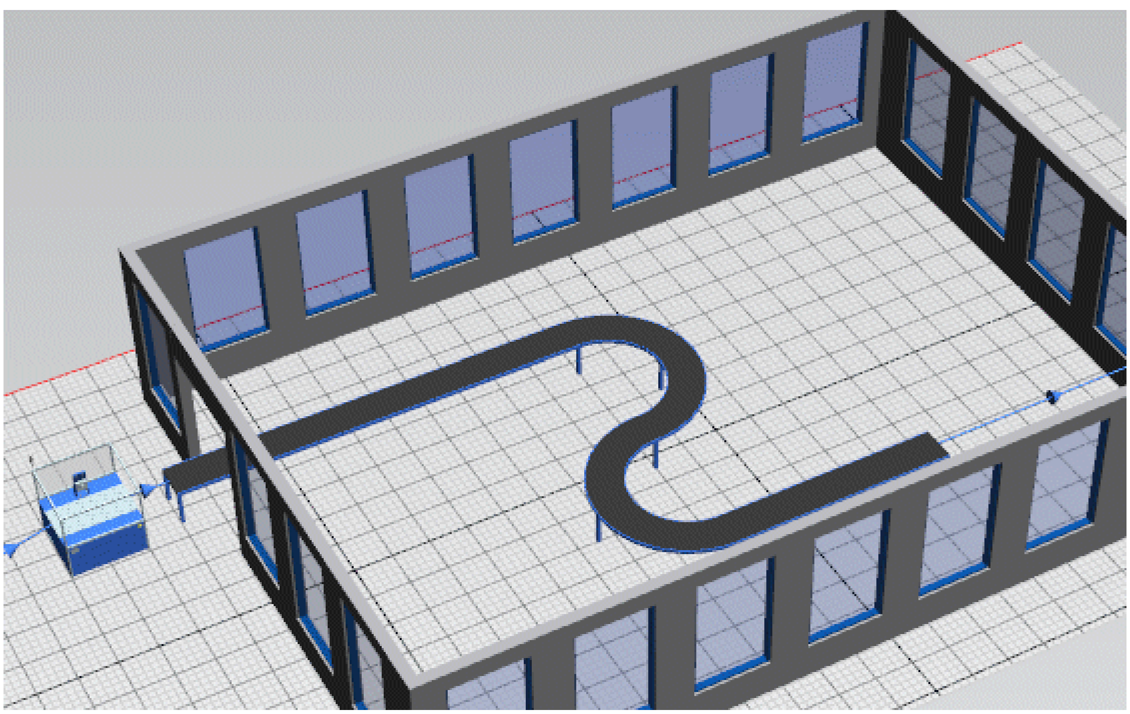
- Open the **3D properties** of the lintel. Change to the tab **Material** and paste the copied material.



- Now we can extend the *conveyor* so that it goes through the passageway.



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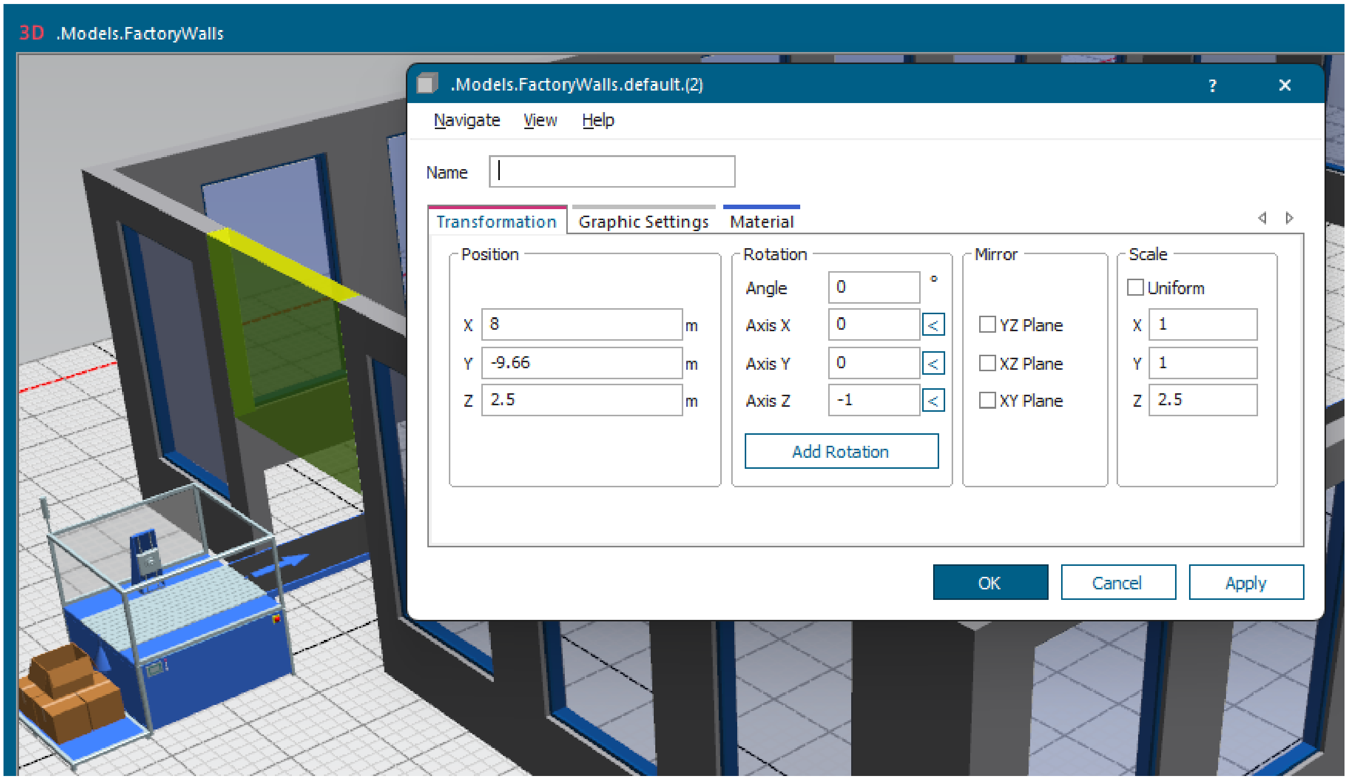
- We can also make the lintel higher by changing the **Scaling** in the **Z dimension**.



Back to Insert Factory Walls and Create Passageways

Create a Passageway and a Gate

Create a Passageway and a Gate

You can create a passageway and a gate through the factory walls.

In the example above we cut an opening into the wall, which might be covered by a PVC strip curtain for through traffic.

Now we create a gate that opens automatically when a part approaches and closes again once it is out of the gate. We model this with a plate that moves up and down.

Proceed as follows:

- Repeat the steps described above to cut the selected segment out of the factory wall.
- Insert a **pillar** (cuboid) each left and right into the passageway and a **lintel** that rests on the pillars.
  - For the two **pillars** we selected these settings:

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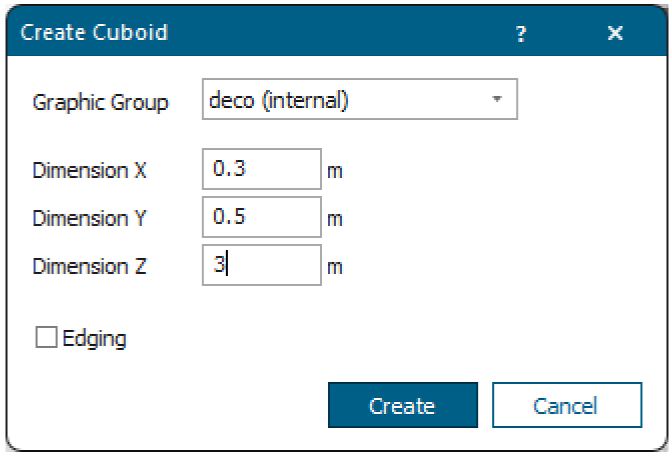
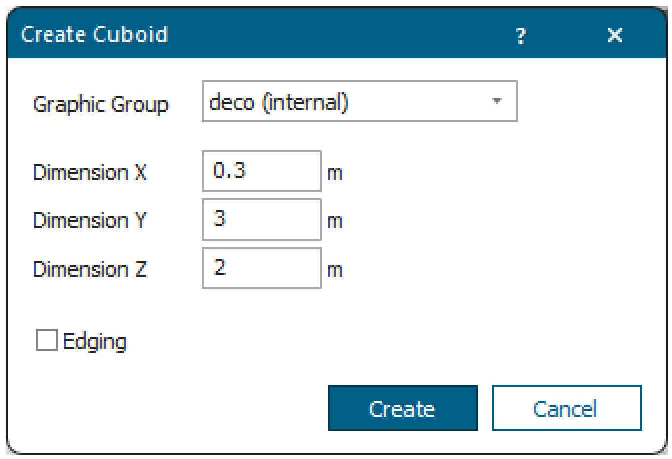
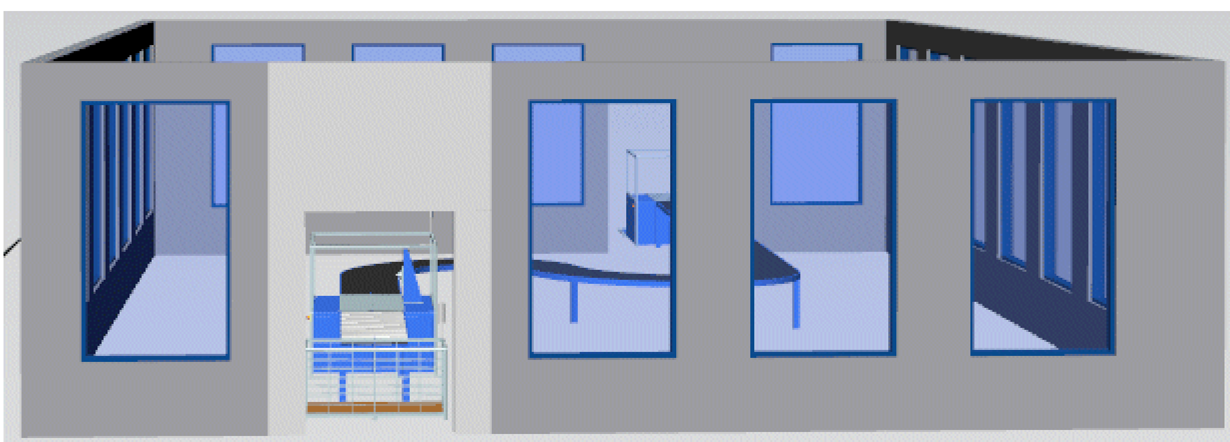


- For the **lintel** we selected these settings:

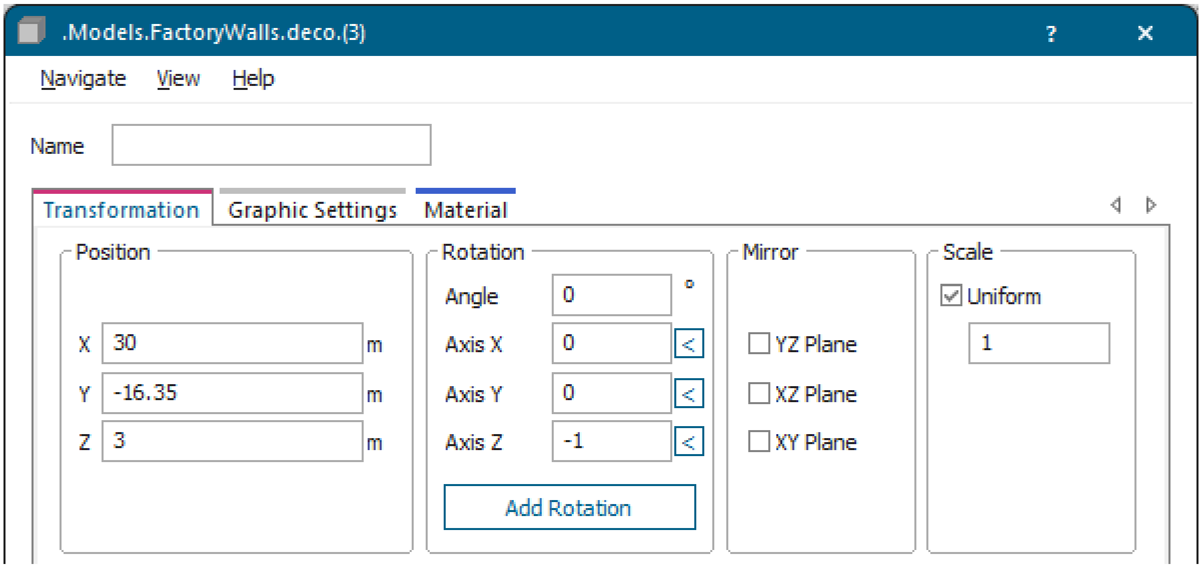


The result looks like this after we adjusted the positions on the tab **Transformation**.



To fine-tune the positions, we clicked in the respective text box, held down **Ctrl** and rolled the mouse wheel. The position of the lintel looks like this.

Unpublished work. © 2026 Siemens



We then create the gate and then can:

- Open and Close the Gate with Poses
- Open and Close the Gate without Poses
- Move the Gate Up to the Height of the Part

Back to Cut an Open Passageway in a Wall

Back to Insert Factory Walls and Create Passageways

Open and Close the Gate with Poses

First we model opening and closing the gate with a joint and with two poses. In addition, we will program three controls.

Proceed as follows:

- Create the graphic for the **gate**. We use a flattened cuboid with these settings.

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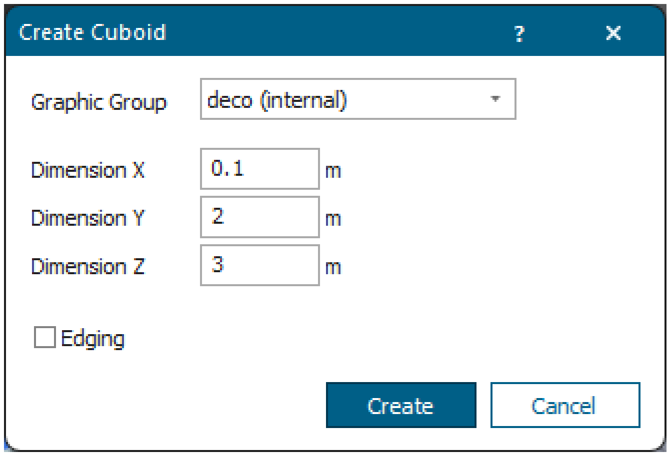
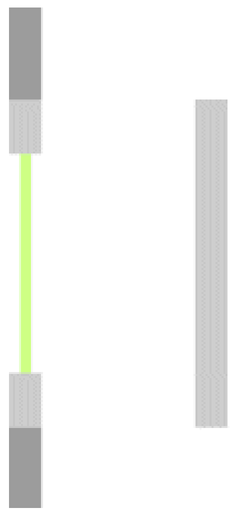
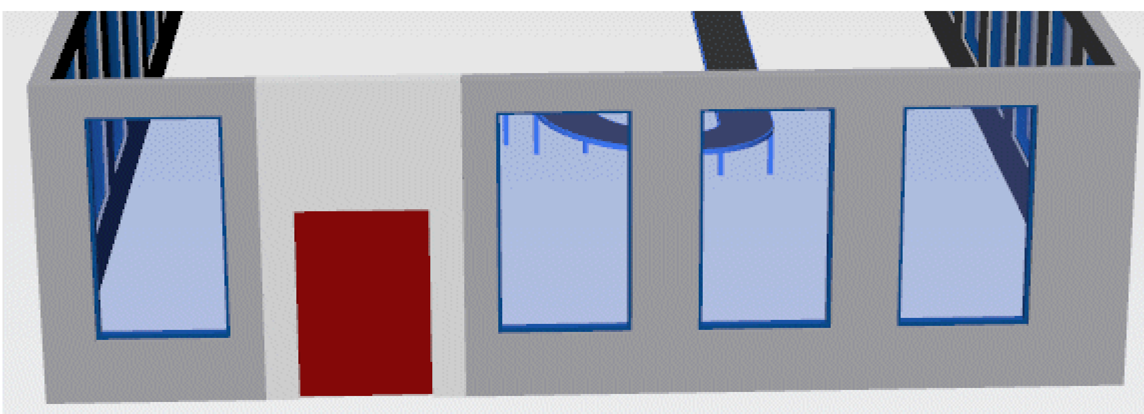


Position the gate between the pillars. To be able to do so, we first moved the lintel to the side.



Now we color the gate so that it stands out from the wall. The result looks like this.



- Then, we insert a *Conveyor* between the *Station* in the factory and the *Drain* outside of it. To overcome the height difference of a meter, we use a spiral conveyor with these settings, compare **Model a Spiral Conveyor**.

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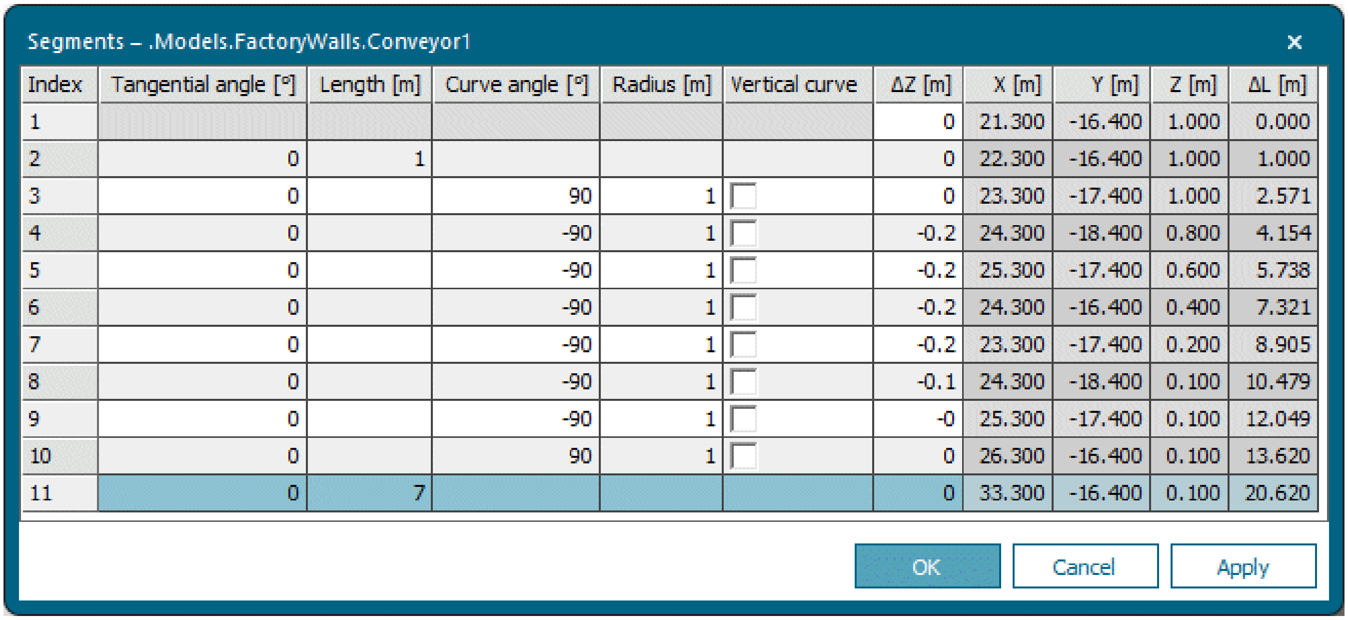


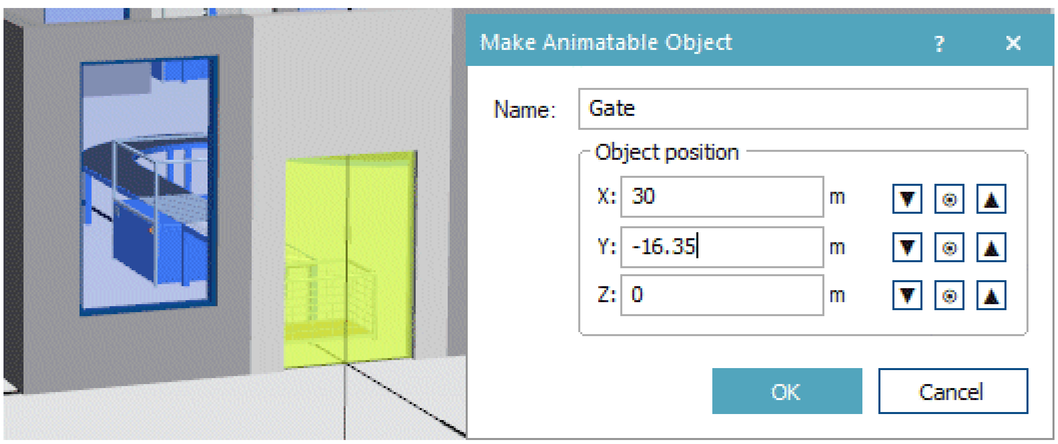
Segments – .Models.FactoryWalls.Conveyor1

Index	Tangential angle [°]	Length [m]	Curve angle [°]	Radius [m]	Vertical curve	ΔZ [m]	X [m]	Y [m]	Z [m]	ΔL [m]
1						0	21.300	-16.400	1.000	0.000
2	0	1				0	22.300	-16.400	1.000	1.000
3	0		90	1	☐	0	23.300	-17.400	1.000	2.571
4	0		-90	1	☐	-0.2	24.300	-18.400	0.800	4.154
5	0		-90	1	☐	-0.2	25.300	-17.400	0.600	5.738
6	0		-90	1	☐	-0.2	24.300	-16.400	0.400	7.321
7	0		-90	1	☐	-0.2	23.300	-17.400	0.200	8.905
8	0		-90	1	☐	-0.1	24.300	-18.400	0.100	10.479
9	0		-90	1	☐	-0	25.300	-17.400	0.100	12.049
10	0		90	1	☐	0	26.300	-16.400	0.100	13.620
11	0	7				0	33.300	-16.400	0.100	20.620

- In the next step we create an object out of the gate that can move up and down.

  Click the gate with the right mouse button and select **Make Animatable Object**.

  Click **OK**.



- Select the joint properties on the **Tab Joint**. We use a **Prismatic Joint** that starts at 0 meters and moves upward to a height of 2 meters. To do so, we selected these settings.

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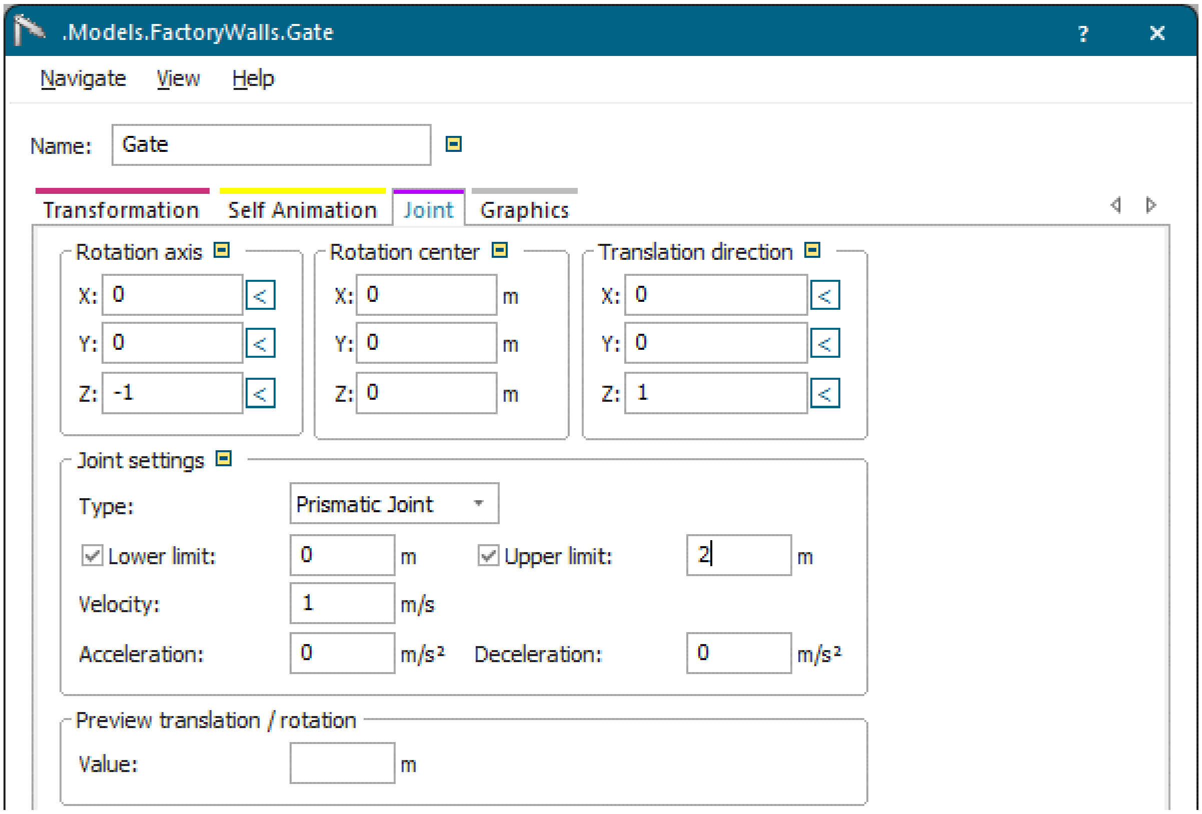
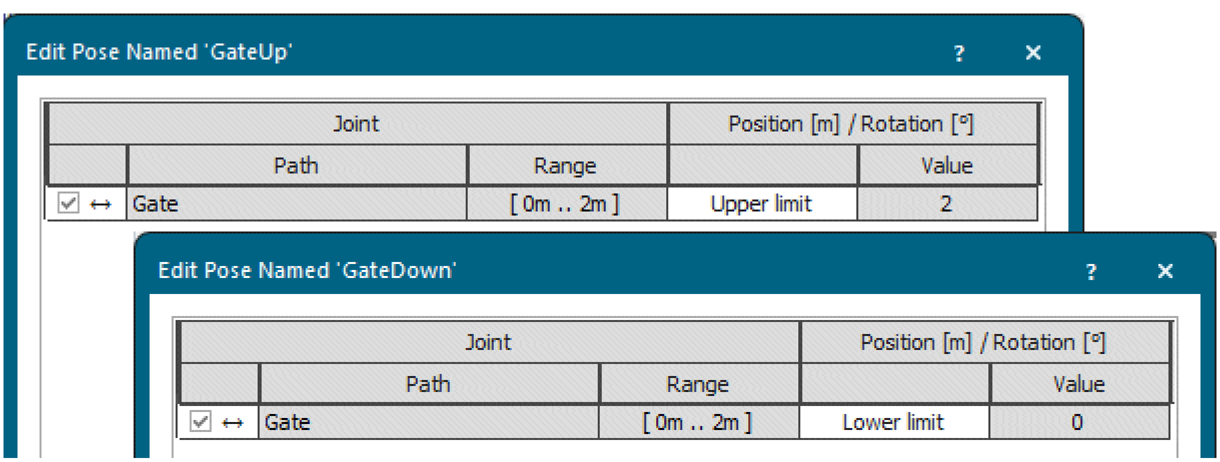


- Click in the background in the *Frame* named **FactoryWalls** and select **Edit 3D Properties**. Change to the **Tab Poses**.

  Add two poses. We named our poses `GateUp` and `GateDown` and selected these settings.

  - **GateUp** moves the gate up to a height of 2 meters.
  - **GateDown** moves the gate down, back to the original position of 0 meters.



- Now we have to set when the gate moves up and down. We do this in an **Entrance Control** and an **Exit Control** of the *conveyor* that transports the parts to the *Drain*.

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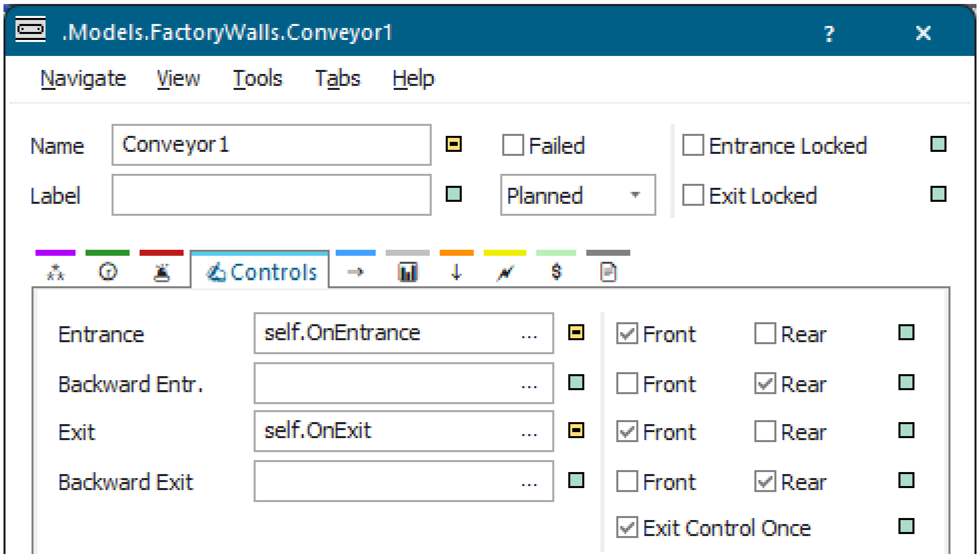


We typed this source code into the **Entrance Control**:

```
self.~.EntranceLocked := true
_3D.Poses.moveTo("GateUp")
self.~.ExitLocked := false
```

We typed this source code into the **Exit Control**:

```
self.~.ExitLocked := true
wait _3D.Poses.moveTo("GateDown")
@.move
self.~.EntranceLocked := false
```

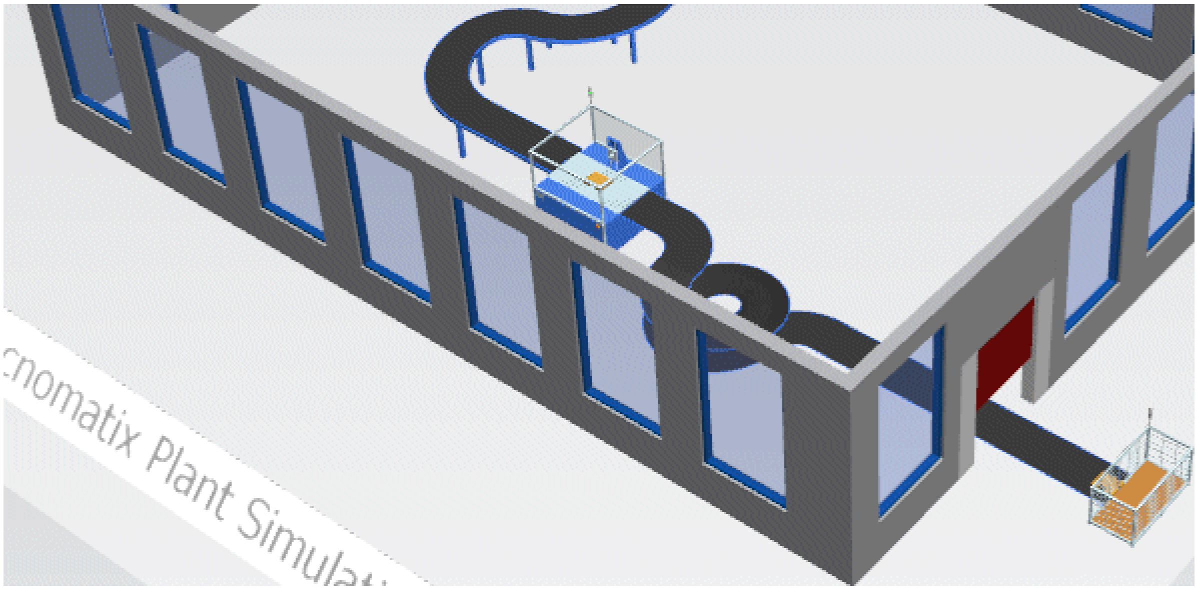
We finally created a *user-defined attribute* and named it `init`. It makes sure that the **exit** of the *conveyor* is closed and that its **entrance** is open when we reset the model.

We typed in this source code:

```
self.~.ExitLocked := true
self.~.EntranceLocked := false
```

- When we now run the simulation, we can watch the gate move up when a part approaches, stays up until the part has passed through the passageway and then moves down again.

Unpublished work. © 2026 Siemens



Back to Create a Passageway and a Gate

Open and Close the Gate without Poses

Here we show how to model opening and closing the gate without poses.

To do so, we have to change the code of our controls and use the method **moveTo**.

Here the gate also moves 2 meters up, stops, and then returns to the initial position of 0 meters after the part passed through the passageway.

- We changed the source code of the **Entrance Control** as follows:

```
self.~.EntranceLocked := true
wait _3D.getObject("Gate").moveTo(2) // gate moves 2 meters up
// wait _3D.Poses.moveTo("GateUp")
self.~.ExitLocked := false
```

- We changed the source code of the **Exit Control** as follows:

```
self.~.ExitLocked := true
wait _3D.getObject("Gate").moveTo(0) // moves the gate back to initial
position
// wait _3D.Poses.moveTo("GateDown")
@.move
self.~.EntranceLocked := false
```

- We did not change the source code of the **Init Control**.

When we run the simulation it looks like this:

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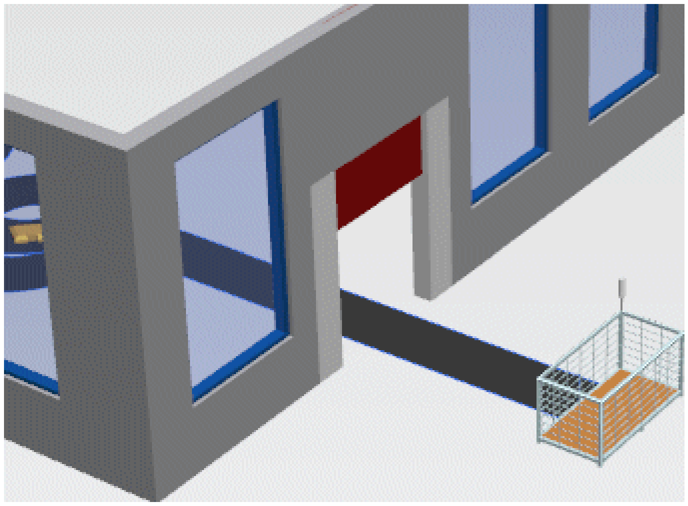

Back to Create a Passageway and a Gate

Move the Gate Up to the Height of the Part

Here we show how to only move the gate up to the height of the part.

To do so, we extended the source code of our methods.

- We changed the source code of the **Entrance Control** as follows:

```
self.~.EntranceLocked := true
wait _3D.getObject("Gate").moveTo(@.MUHeight+0.1m)
// moves the gate up to the height of the part
// plus 10 cm for the height of the conveyor
wait _3D.Poses.moveTo("GateUp")
self.~.ExitLocked := false
```

- We changed the source code of the **Exit Control** as follows:

```
self.~.ExitLocked := true
wait _3D.getObject("Gate").moveTo(0.1m)
// moves the gate back to the initial position
wait _3D.Poses.moveTo("GateDown")
@.move
self.~.EntranceLocked := false
```

- We changed the source code of the **Init Control** as follows:

```
self.~.ExitLocked := true
self.~.EntranceLocked := false
_3D.getObject("Gate").moveTo(0.1m, 0)
// 0 means that the target position becomes
// active immediately without moving there
```


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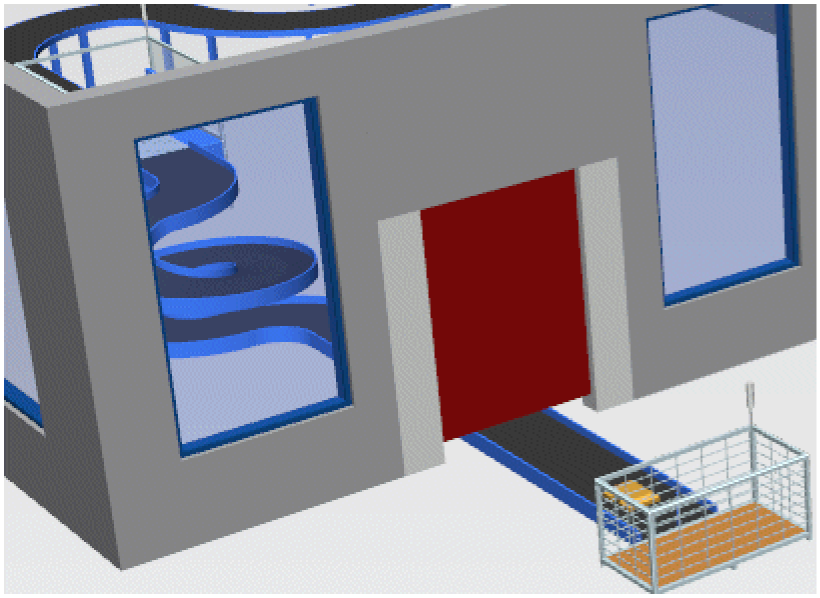
When we run the simulation it looks like this:



Back to Create a Passageway and a Gate

Adding Text and Display Boards

Add Text and Display Boards

Sometimes you might want to show text and display boards in your model to make it visually pleasing and to present information to the user.

- If you want to show **static text** that you do not need to change after creating it, you can insert **3D Text**.

  You can also import a JT file, which you, for example, created in *NX*. We created the lettering **Plant Simulation** in the sample model **Factory 51** like that.

- If you want to create a **display board**, which you can edit after creating it, use the object Comment.

You can:

- Show Text
- Show a Display Board

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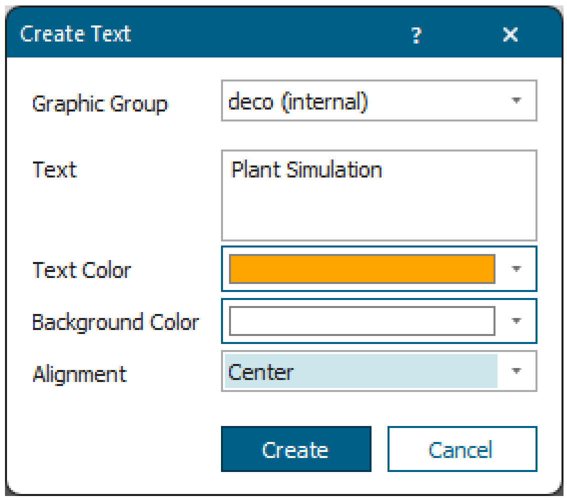
Back to **Create a Realistic Looking Model**

Show Text

To insert static text into your simulation model, click **Text** under **Edit > Insert Shape**.

- Type in the text that you want to show and select the settings in the dialog. Click **Create**.



- Drag the mouse to a position of your choice and click the left mouse button. This inserts the text flat on the floor of the factory. As compared to the factory building it is much too small though.

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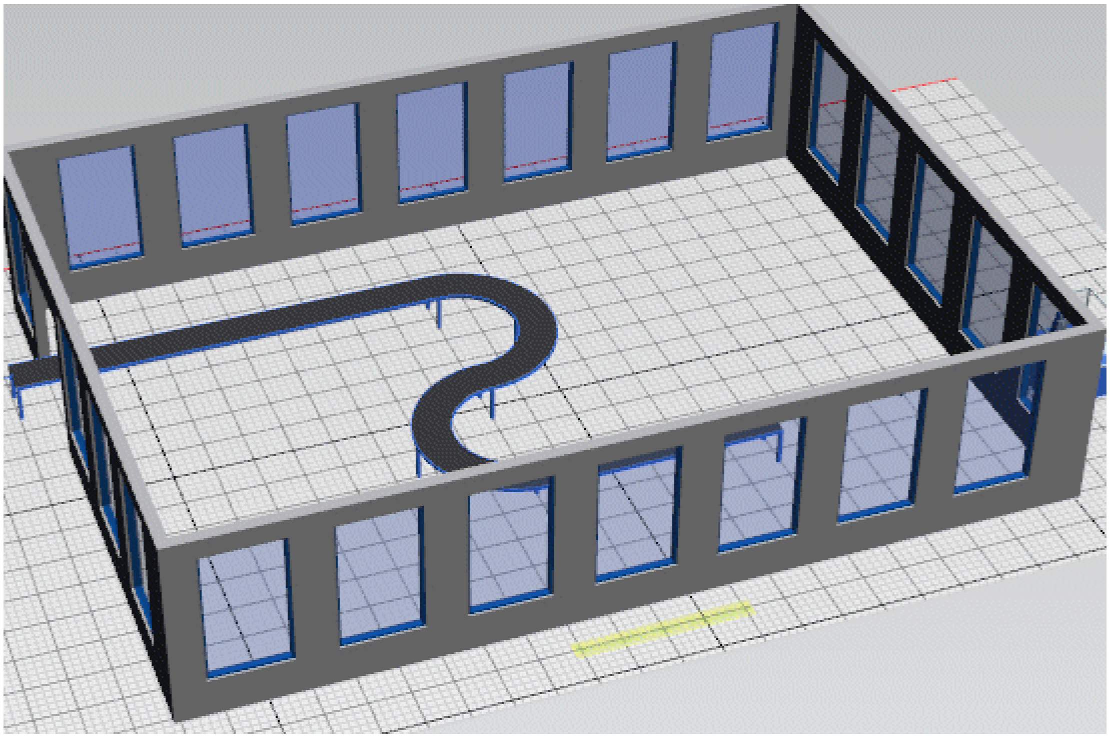
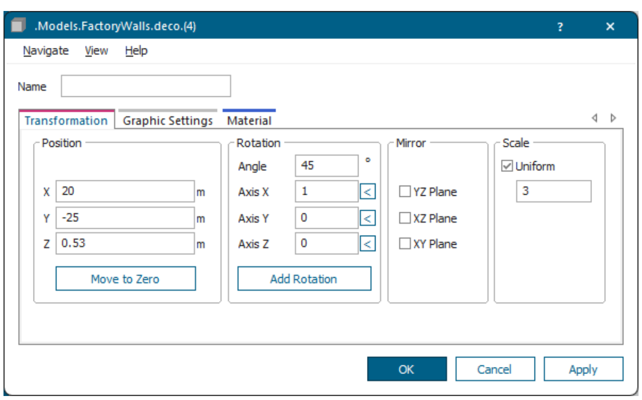
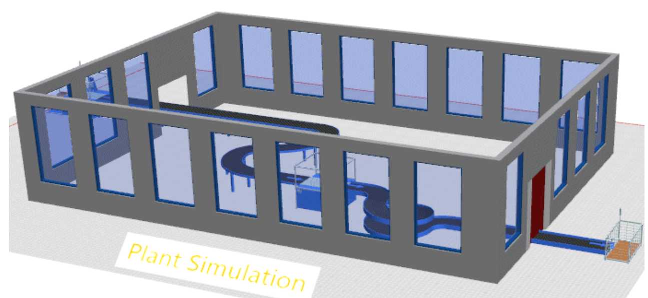


- To fix this, we adjust the size and the position of the lettering so that it matches the scale of the factory building. Click the text with the right mouse button and select **Edit 3D Properties**.
  - We selected **Uniform Scaling** with a factor of 3.
  - We rotated the text by 45 degrees around the **X** axis (1).
  - We moved it up on the **Z** axis until the lower border is located flat on the floor.

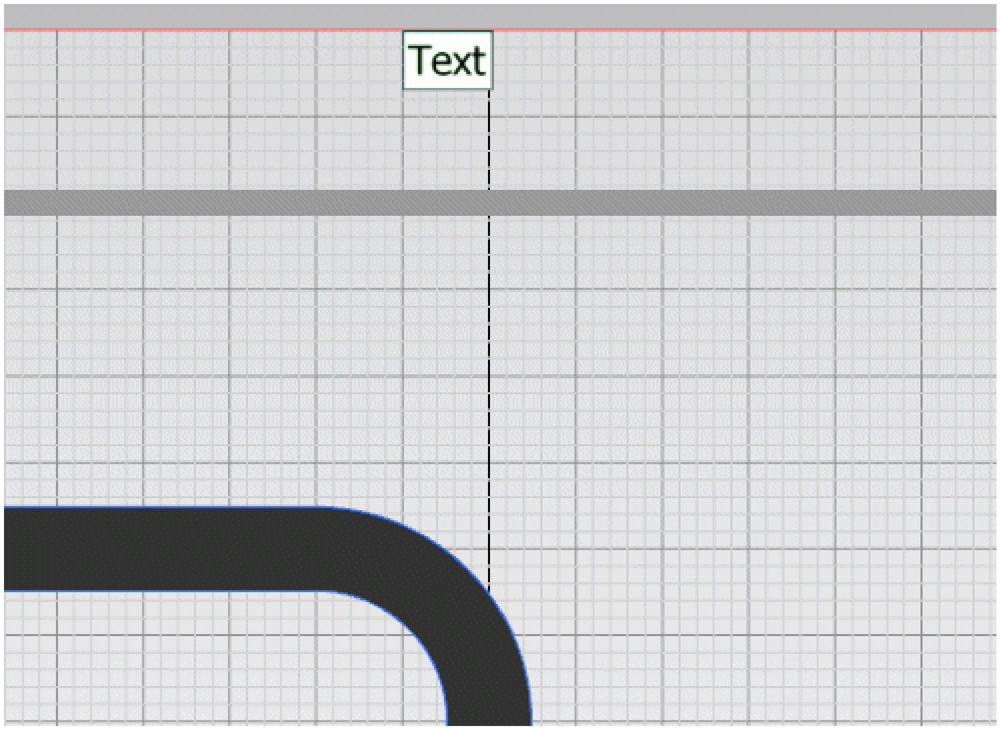




Back to Add Text and Display Boards

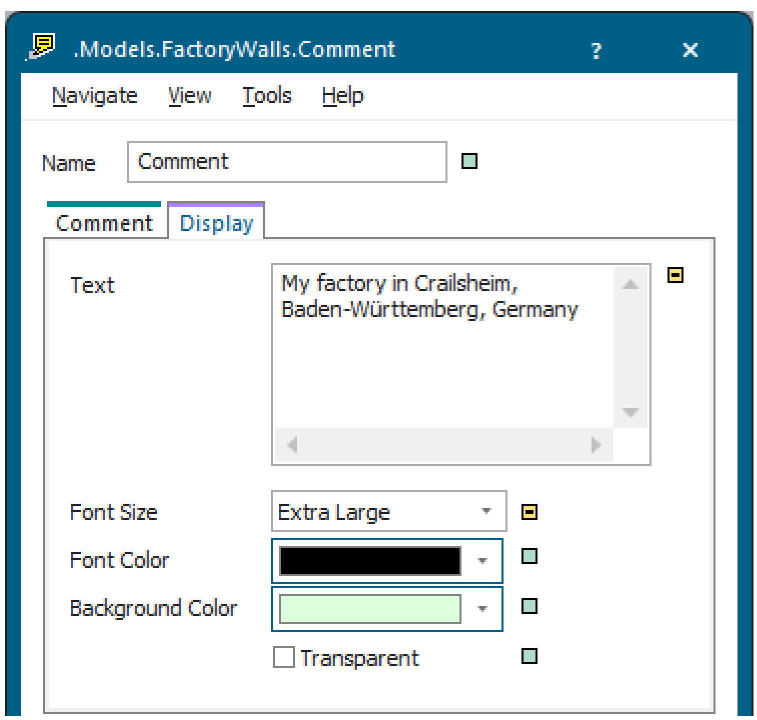
Show a Display Board

To show a display board, click the object *Comment* on the toolbar **User Interface** and insert it into your model.



- Double-click the *Comment* and type in the text that you would like to show.

  We typed in `My factory in Crailsheim, Baden-Württemberg, Germany` and selected the **Font Size > Extra Large**.



- Click **OK**.

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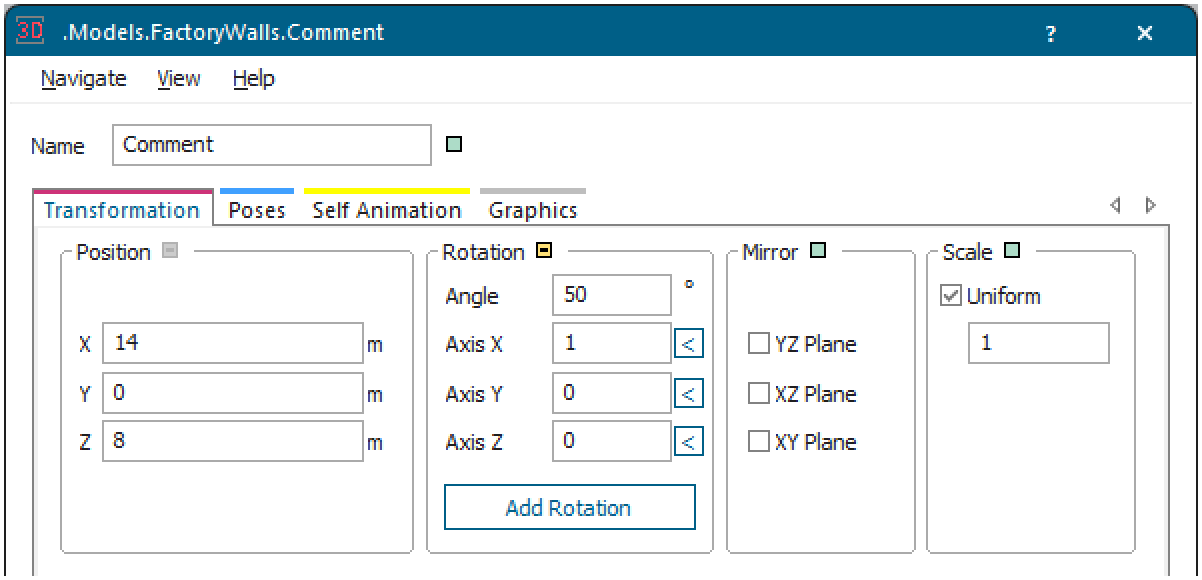
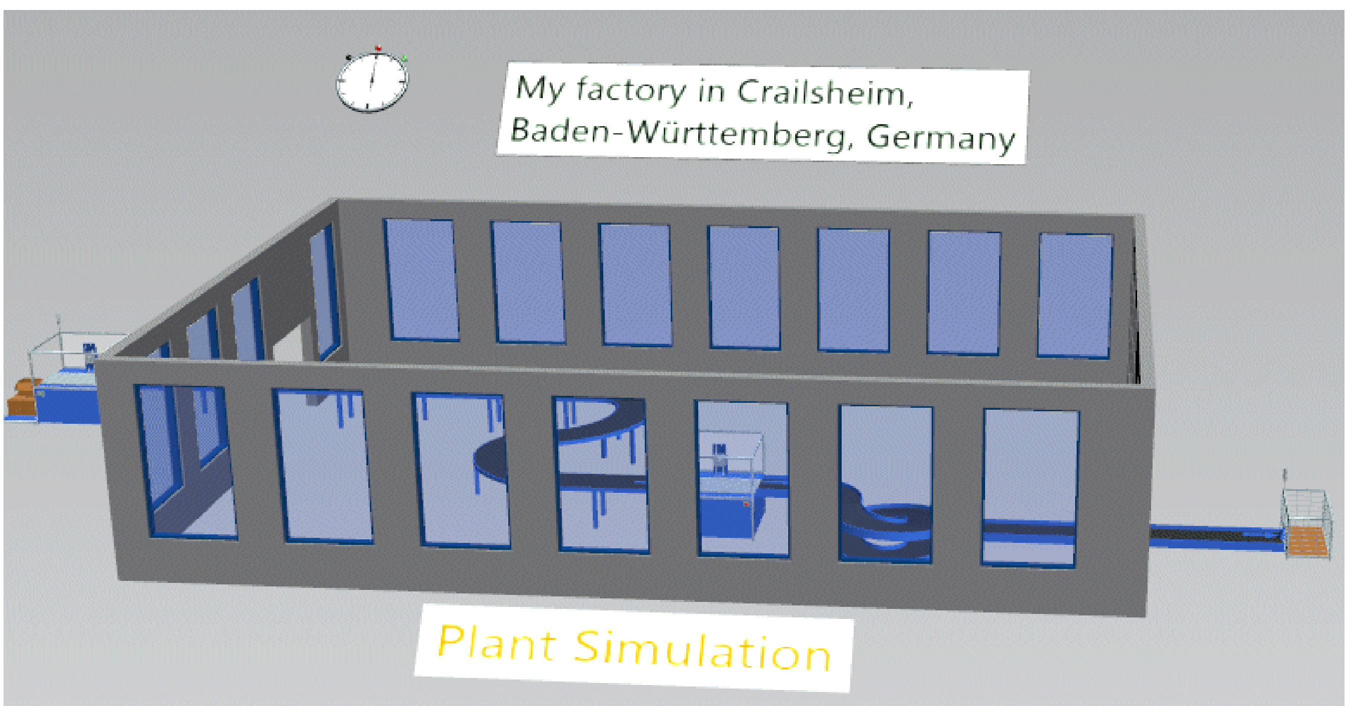
- We want to show the display board rotated to the factory floor and above the upper border of the hall though.

  Click the display board with the right mouse button and select **Edit 3D Properties**.

  - Rotate the display board by 45 degrees around the **X** axis.
  - Move the position of display board up on the **Z** axis. We selected these settings.



- Click **OK**. The result looks like this:



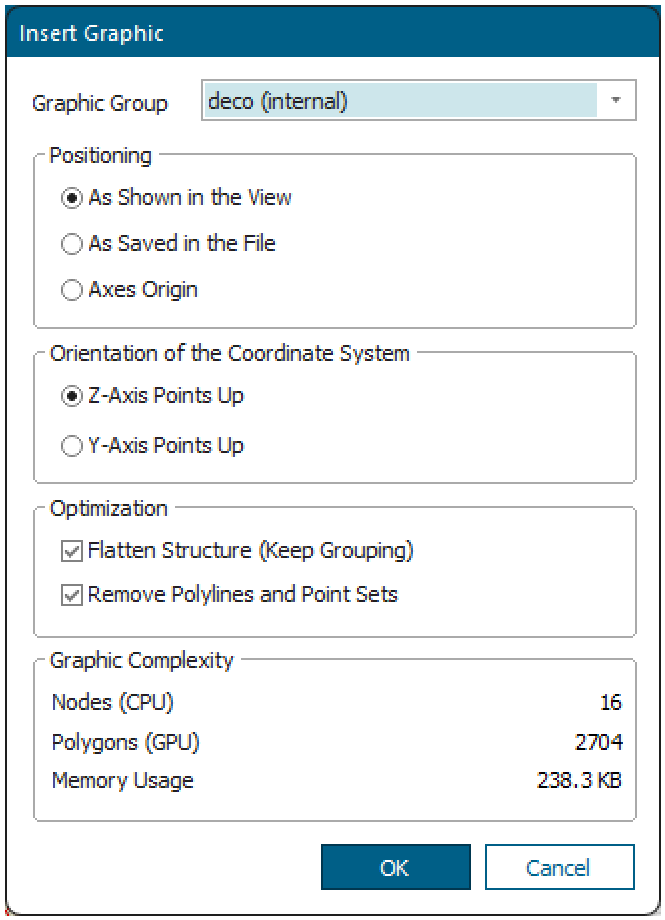
Back to Add Text and Display Boards

Add a JT Layout File to Your Simulation Model

A layout graphic adds a realistic touch to your simulation model.

To add a JT graphic to your model:

- Drag the layout graphic, *Layout_Factory_Training.jt* in our case, which we received from a colleague, from the Windows Explorer to the open *Frame* window containing the model and drop it there. *Plant Simulation* attaches the graphic to the mouse pointer and opens the dialog **Insert Graphic**. Accept the default settings and click **OK**.



- Drag the mouse to the upper left corner of the *Frame* to the coordinates **0**, **0** and click the left mouse button to place and insert the graphic. The model should look like this now:

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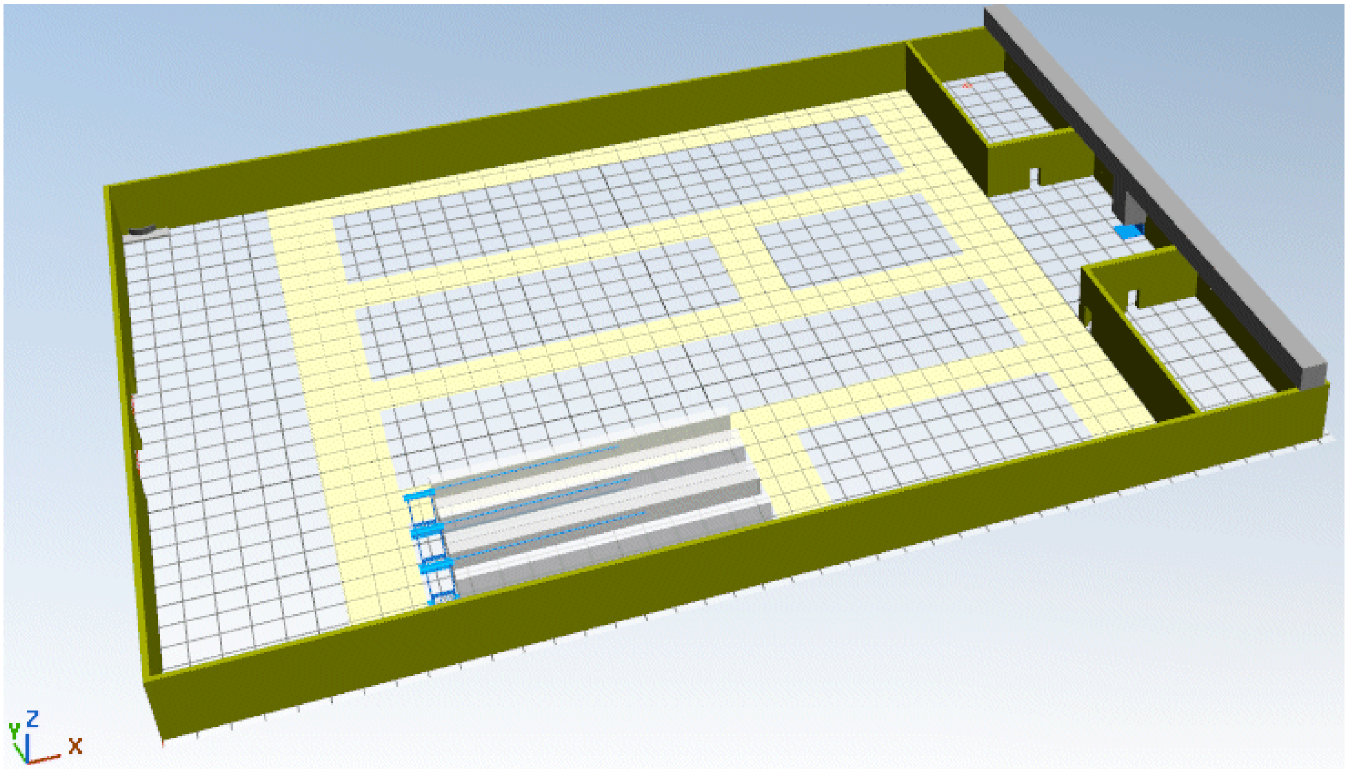

You can now continue working with your model, by inserting objects and components and by manipulating them, etc.

Instead of using a *JT-layout file* you can also use a *point cloud* as the background of your simulation model.

Compare Import a 3D Graphic

Compare Add a Background Graphic to the Frame

Back to Create a Realistic Looking Model

Add a Point Cloud

A point cloud is a set of data points in three-dimensional space. These points are defined by their X-, Y-, and Z-coordinates. You might use a point cloud to visualize your actual factory layout and you can arrange your machines on the graphic to give it a realistic touch.

To import a point cloud into your simulation model:

- Click into the background of your simulation model and select **Edit 3D Properties**. Change to the Tab Point Cloud.
- Click the button and select the point database file (*.**pod**) that you want to import. If the path is relative, it refers to the folder in which your simulation model is located.

  To remove the point cloud from the simulation model, delete its name from the text box.

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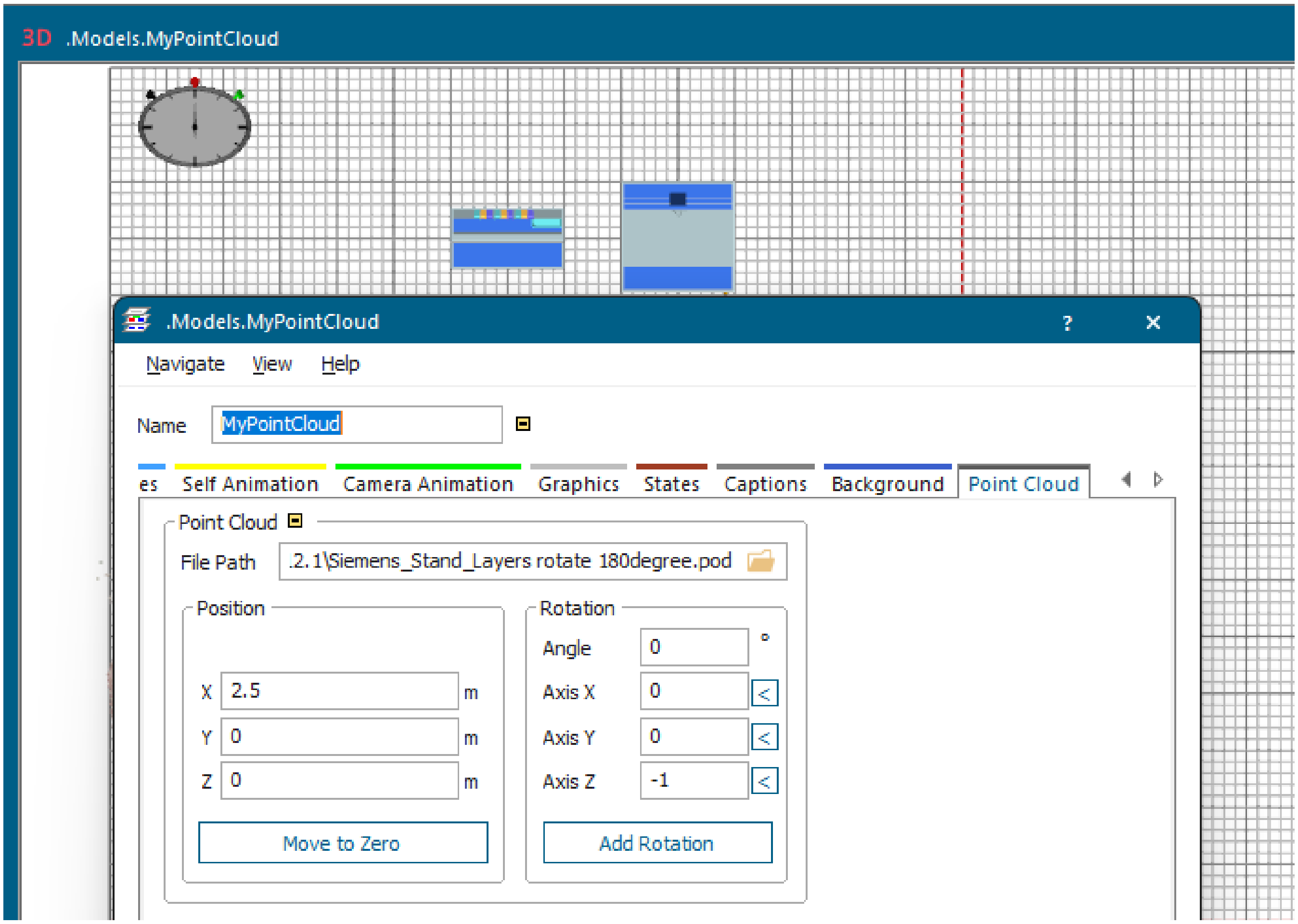


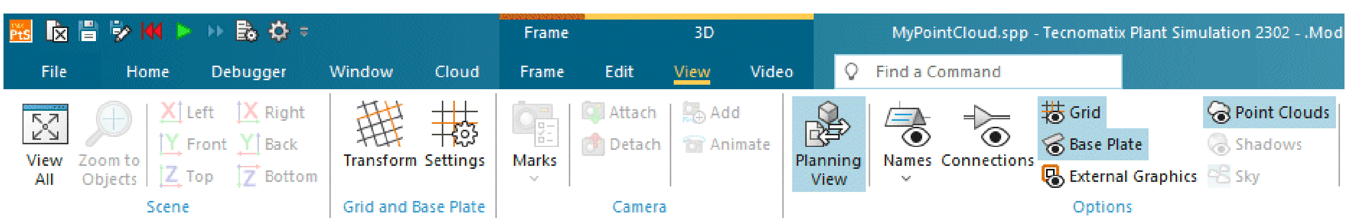
- Click **OK** or **Apply** to show the point cloud.

Note

Initially, your window might only show small parts of the point cloud. When you start rotating the view, *3D* shows additional parts of the graphic.

Plant Simulation will load the point cloud significantly faster if the point cloud file is located on an SSD storage device.

- To show or hide the point cloud, click **Show Point Clouds** on the **View** ribbon tab.



- To adjust the position and the orientation of your point cloud, click into the background of the *Frame* containing the point cloud and select **Edit 3D Properties**. Change to the tab **Point Cloud** and modify its **Position** and its **Rotation Angle**.

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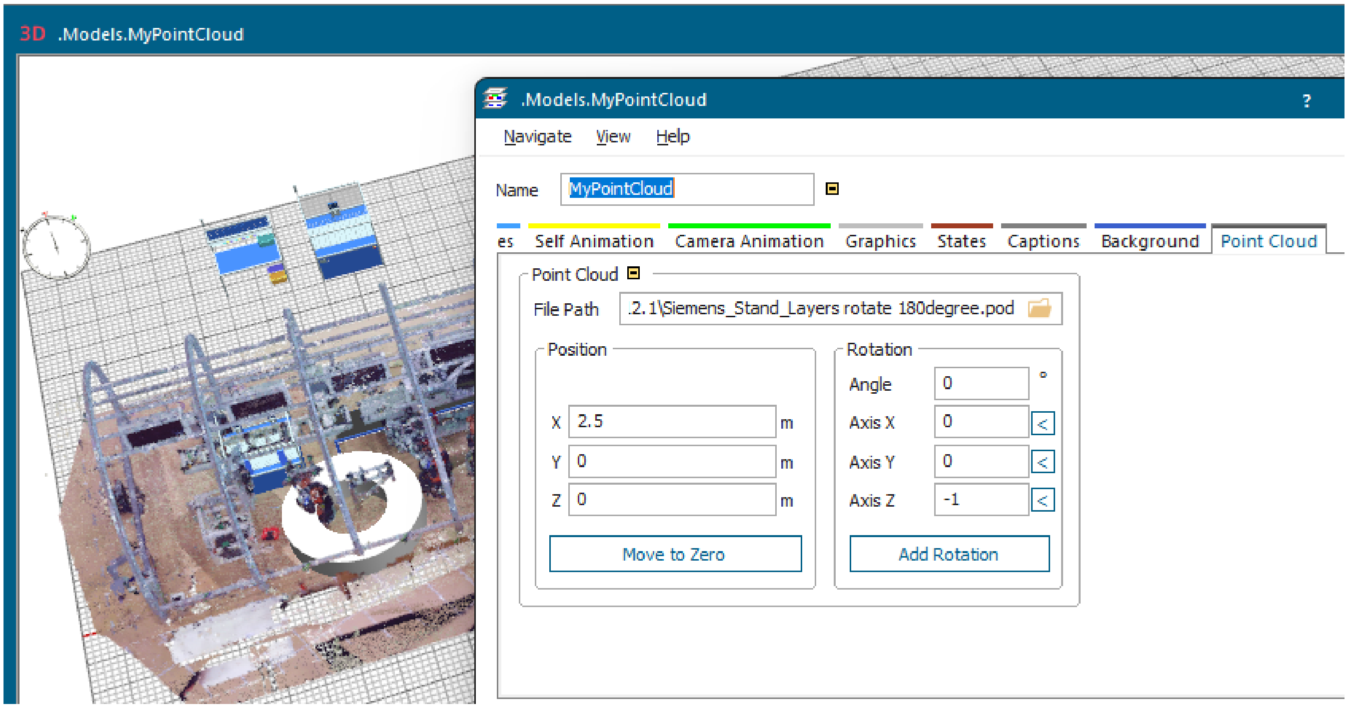
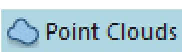


- Insert the objects of your simulation model at their respective positions.

Be aware that for performance reasons **Show Point Clouds** is deactivated by default. For this reason you will not see the point cloud when you reopen the model at a later point in time. Click **Show Point Clouds** 

to display it.

Back to Create a Realistic Looking Model

Import JT Graphics Representing an Object

Instead of using shapes, which you create in *3D* itself, you can also import graphics to represent an object. In our example we used a number of JT graphics, which we received from a colleague, to build a dune buggy.

Proceed as follows:

- Duplicate the part **.MUs.Part** in the *Class Library* and rename it to DuneBuggy. Add the graphic groups containing the graphics of the components on the Tab Graphics. We added the following graphic groups and named them as shown in the screenshot below.

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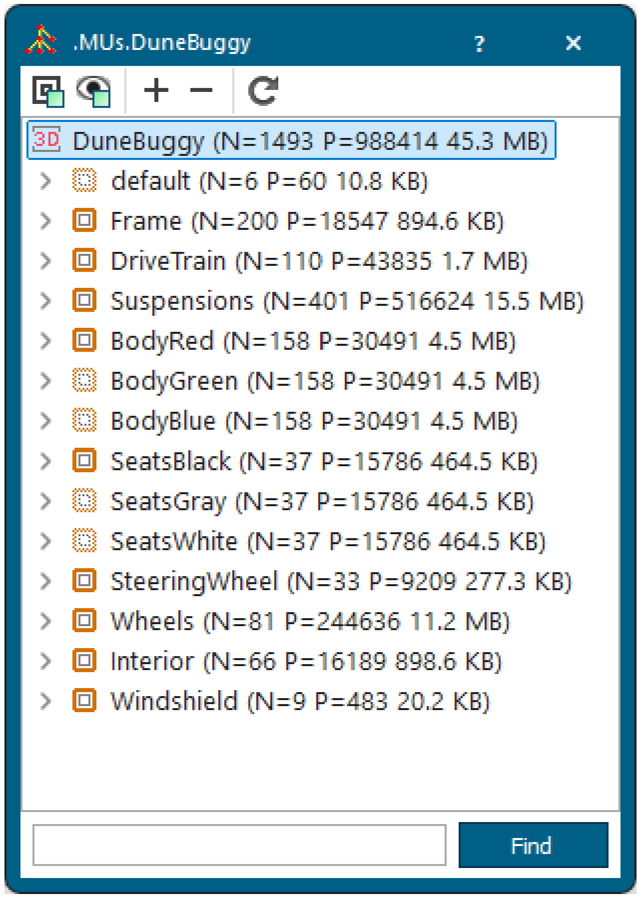

- Import the JT graphics into the graphic groups with the respective name. Import the files *Body.jt* and *Seats.jt* into all graphic groups whose name starts with **Body** or **Seats**.

Name	Date modified	Type	Size
Body.jt	23.02.2015 15:20	JT File	597 KB
DriveTrain.jt	23.02.2015 14:05	JT File	712 KB
Frame.jt	11.10.2007 16:52	JT File	497 KB
Interior.jt	23.02.2015 14:43	JT File	219 KB
Seats.jt	23.02.2015 14:54	JT File	191 KB
SteeringWheel.jt	23.02.2015 14:45	JT File	125 KB
Suspensions.jt	23.02.2015 14:04	JT File	7.218 KB
Wheels.jt	23.02.2015 14:56	JT File	2.383 KB
Windshield.jt	23.02.2015 15:24	JT File	13 KB

  - Change to the 3D ribbon tab **Edit** and click **Import Graphics**, navigate to the folder containing the JT graphics, and click **Open** in the dialog **Open**.

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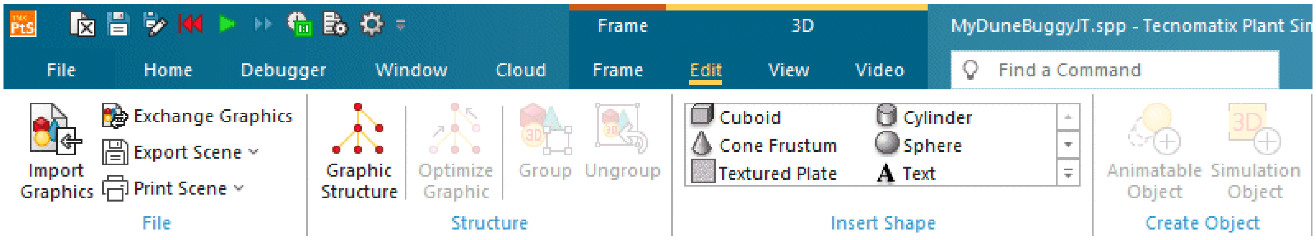
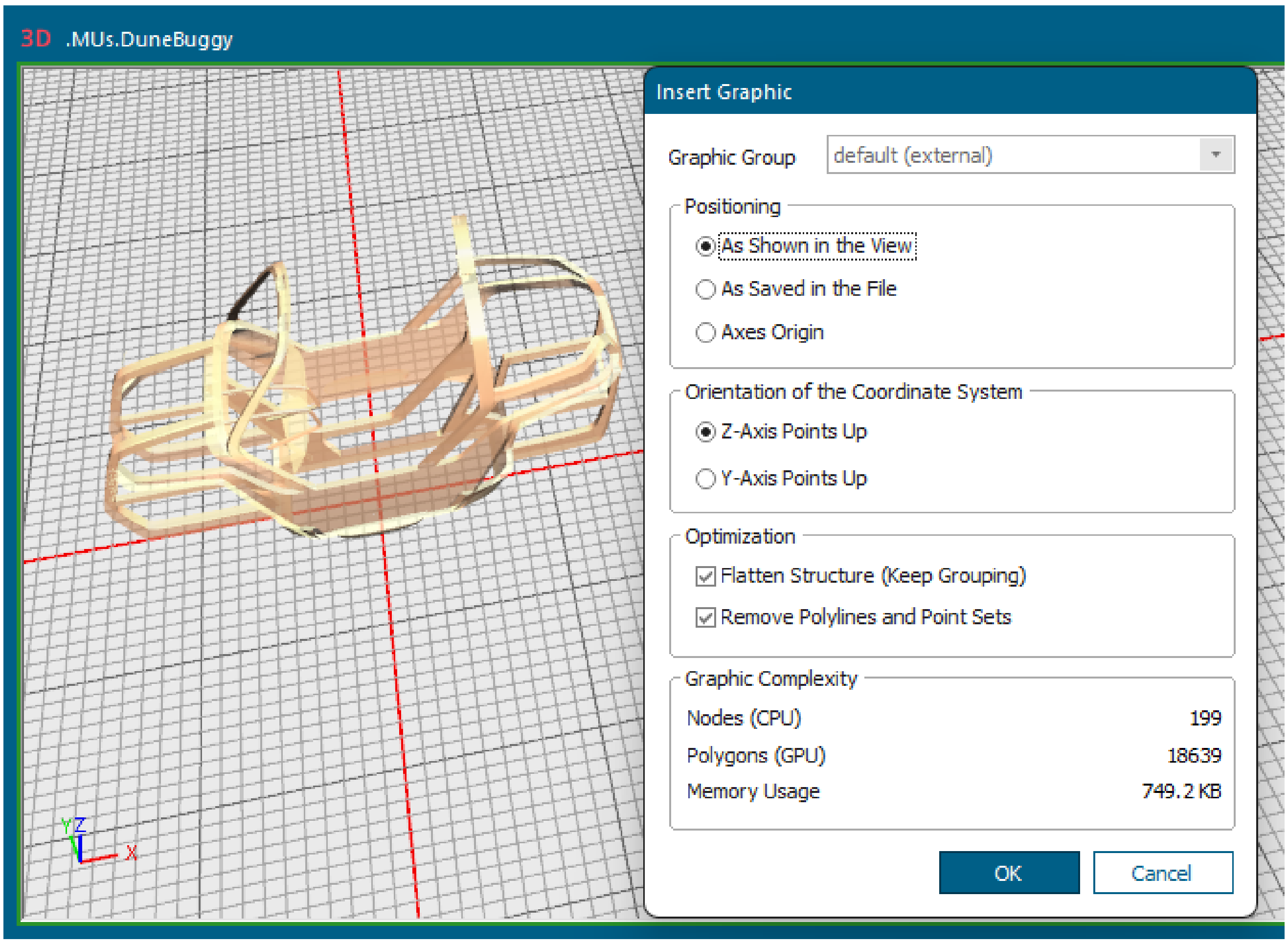


- Drag the JT graphic, the **Frame** of the dune buggy in the example below, roughly to the position in the *Frame* named **.MUs.DuneBuggy**, where you want it to be, and click the left mouse button. Select the graphic group to which you want to add the graphic, **Frame** in our case.



- Repeat the steps above for all JT files.
- Arrange the position of the components of the dune buggy. We used a combination of the arrow keys and the settings in the dialog **Edit 3D Properties**.

  Place the three body panels at exactly the same position. Do the same for the three seat assemblies. This way you can later show or hide selected body panels and seats to test different color combinations.

- To change the color of a graphic, click into the background of the *Frame* named **.MUs.DuneBuggy** with the right mouse button, and select **Show Graphic Structure**.

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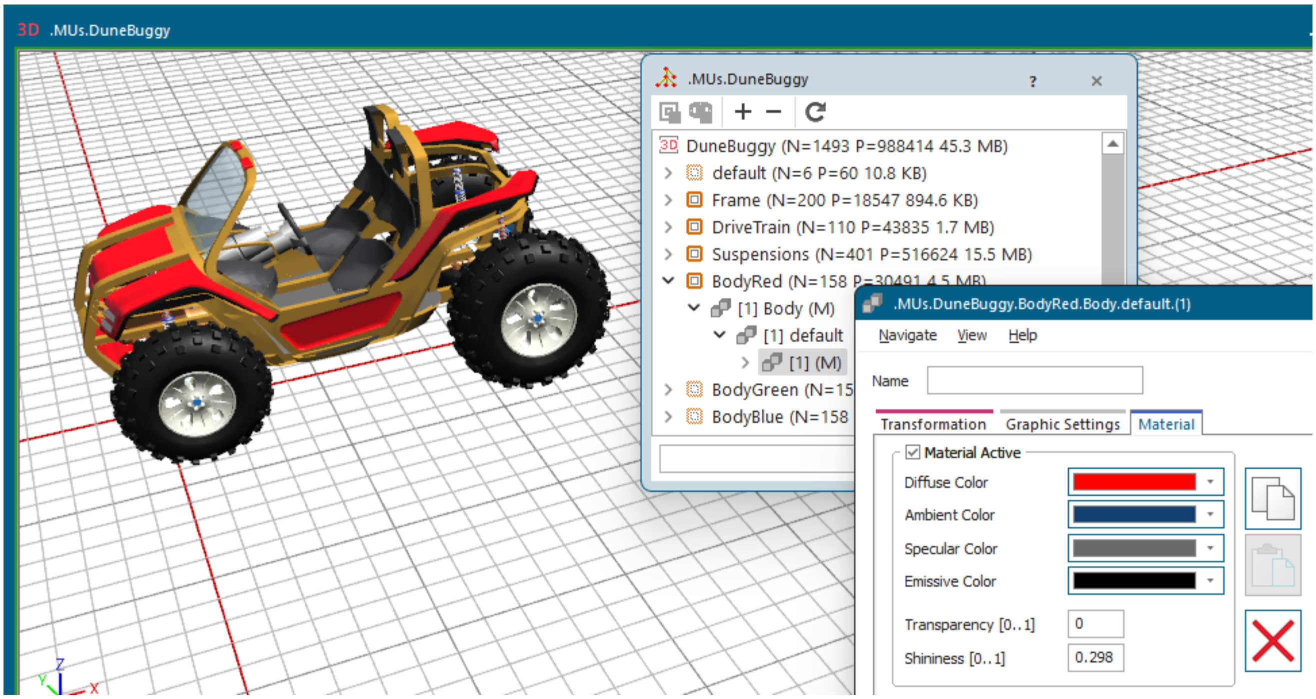
- Expand the graphic group by clicking the ⊞ button, **BodyRed** in our case.
- Look for the node with a **-M** at the end of the name. This designates that this node defines a material.
- Click that node with the right mouse button, **(1-JtGroup-M)** in our case, and select **Edit 3D Properties**. Experiment with the different material colors to create the look that you want. Using the color combinations, you can fine-tune glossy effects etc.



Back to **Create a Realistic Looking Model**

Optimizing a Graphic

Optimizing the graphic, i.e., flattening the hierarchy of an object, deletes unnecessary information from the object, allowing for faster rendering, and a quicker display of the object.

Whenever you import graphics, it is useful to optimize them. This is not because the graphics you import are badly designed, but because they are typically designed for another kind of use in the originating program. The engineer who designs a component, such as an engine, will naturally include every detail of that engine in the graphic that he creates, because it is imperative to provide these details for producing the engine.

For us as users these details of the engine are of no interest at all. Our main concern is that the simulation runs fast and that the animation looks good and runs smoothly.

We therefore advise to optimize every graphic you import. Do this as soon as possible, but not before you extracted all the structural information from the graphic that you need, compare **Create a Simulation Object with Animation**.

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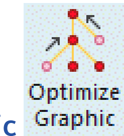
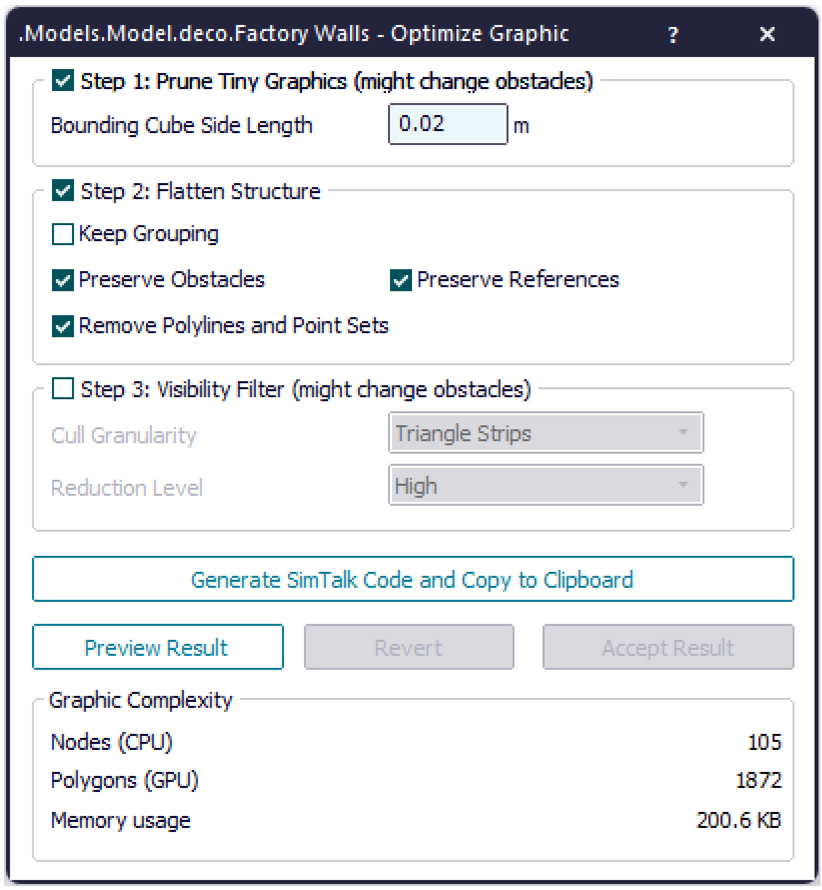
Before you can do this, you have to make an animatable object from the graphic, compare **Make Animatable Object**.

To optimize a 3D graphic, click the animatable object, whose graphic you would like to optimize, in the scene window, and click **Optimize Selected Graphic** 

on the **Edit** ribbon tab of the *3D window*. Experiment with the settings described under **Step 1: Prune Tiny Graphics**, **Step 2: Flatten Structure**, and **Step 3: Visibility Filter**.



Note

Click **Show External Graphic Group**  on the **View** ribbon tab to display the external graphics of the open 3D object.

Plant Simulation shows two characteristics of the graphic under **Graphic Complexity**, namely the number of nodes and the number of polygons. The optimization strategies mainly affect the number of nodes and the number of polygons.

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